The Spectral Shortcut Theorem: A Gradient-Geometric Explanation for Failure Modes of Joint Spectral-Spatial Deep Learning

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Abstract

[TODO: Write abstract last, after Phase 7.]

1 Introduction

The integration of spectroscopic imaging with deep learning has produced a class of compositional models that has become the de facto standard across infrared tissue classification, hyperspectral remote sensing, Raman imaging, and mass spectrometry imaging. A canonical pipeline processes a hyperspectral input $X \in \mathbb{R}^{H\times W\times S}$ in two stages: a per-pixel spectral reduction $f_{\boldsymbol{\theta}} : \mathbb{R}^S \to \mathbb{R}^K$ followed by a spatial model $g_{\boldsymbol{\phi}} : \mathbb{R}^{K\times H\times W} \to \mathbb{R}^{N_{\mathrm{cls}}\times H\times W}$. End-to-end joint training of this pipeline is overwhelmingly the default — adopted by foundational work in FTIR histology Berisha et al. [2019], recent state-of-the-art spatial-spectral ViTs for prostate tissue classification O'Leary et al. [2026], and essentially all hyperspectral remote sensing systems.

An empirical paradox. Despite the prevalence of this paradigm, an unexpected and persistent pattern has emerged across multiple subfields. Frozen pretrained spectral encoders dramatically outperform end-to-end joint training on held-out test data. In our motivating tissue classification setting, a learned-from-scratch linear $f_{\boldsymbol{\theta}}$ trained end-to-end achieves 0.675 test F1; freezing a pretrained $f_{\boldsymbol{\theta}}$ yields 0.84–0.95 test F1. Most strikingly, replacing $f_{\boldsymbol{\theta}}$ with *frozen random weights* yields 0.70 test F1 — outperforming the joint trained model. Recent independent observations O'Leary et al. [2026], Berisha et al. [2019] have noted, in different terms, that aggressive spectral compression to 16 features has negligible impact on spatial-spectral neural network performance, and have concluded that "tissue classification only needs a small set of spectral features."

We argue the opposite. The reason end-to-end joint training fails is not that spectral information is unimportant. It is that *end-to-end optimization fails to extract that information*, even when the spectral signal is rich and discriminative. The problem is in the training procedure, not the data.

This paper. We provide a rigorous explanation for the failure of joint spectral-spatial training, grounded in the geometry of the loss landscape and the dynamics of stochastic gradient descent on compositional architectures with asymmetric capacity. Two main theorems form our contribution.

- **Theorem 13 (Capacity-Induced Module Curvature Disparity).** We prove that the ratio of the spatial to the spectral Gauss-Newton block's top eigenvalue grows linearly with the spatial module's width M (equivalently, as $\sqrt{C_g/C_f}$ for architecture families with $C_g = \Theta(M^2)$): as the spatial module widens, curvature concentrates in it while the spectral block's curvature stays capped by a width-independent constant. The proof is elementary: an explicit witness direction in the classifier-head weights forces the width-linear growth, and an operator-norm bound caps the spectral block; the rate matches the mean-field law of Karakida et al. Karakida et al. [2019] in the model class where the latter applies.

- **Theorem 28 (Finite-Time Spectral Starvation).** Under an explicit shortcut-fitting hypothesis compatible with cross-entropy dynamics, the spectral module's displacement from initialization by the time the training residual reaches level δ is at most $\mathcal{O}(\varepsilon \log(1/\delta))$, where ε is the ratio of the (capped) spectral gradient scale to the shortcut fitting rate. As spatial capacity grows, the shortcut fits faster while the spectral gradient stays capped, so the spectral module is pinned near initialization for the entire useful duration of training. The formulation is finite-time by design: for unregularized cross-entropy on separable data no finite stationary point exists Soudry et al. [2018], and the starvation phenomenon is a training-time effect, not an asymptotic one.

A real-time diagnostic. The pathology described by Theorems 13 and 28 is invisible to standard loss curves — training and validation loss both decay smoothly while the spectral module is silently starved. To make the pathology observable in practice we introduce the *Effective Gradient Ratio* (EGR), a scalar diagnostic computable inside any training loop:

$$\mathrm{EGR}(t) = \frac{\|\nabla_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}_t, \boldsymbol{\phi}_t)\|}{\|\nabla_{\boldsymbol{\phi}} \mathcal{L}(\boldsymbol{\theta}_t, \boldsymbol{\phi}_t)\|}.$$

Proposition 30 establishes that the spectral gradient norm—the EGR's numerator—decays polynomially while the loss is being fitted. Empirically, the depth of the EGR collapse scales monotonically with capacity ratio, providing a quantitative, training-time observable for gradient starvation.

Prescription. The theory yields a direct prescription: freeze the spectral module, or inject a fixed residual baseline. Freezing eliminates the timescale gap by removing $\boldsymbol{\theta}$ from the optimization entirely. A residual baseline ($Z = f_{\boldsymbol{\theta}}(X) + h(X)$ for a fixed h, e.g. PCA) is a softer intervention that breaks the shortcut without precluding spectral fine-tuning. We discuss both in Section 9.

Universality. Although our motivating application is infrared tissue classification, the mechanism we identify applies to any compositional architecture with capacity asymmetry between sequential modules. We verify in Section 7 that the same pattern manifests across linear, CNN, and Vision Transformer spatial models on synthetic hyperspectral problems, indicating the phenomenon is architecture- agnostic. We expect the same mechanism to govern frozen-encoder practices in Raman spectroscopy, mass spectrometry imaging, NIR analysis, and beyond.

Relation to prior work. The closest prior work is in multimodal learning, where Wang et al. Wang et al. [2020] and Peng et al. Peng et al. [2022] identified "modality competition" and proposed gradient blending and modulation interventions. Our setting is unimodal (a single hyperspectral input) but the mechanism is structurally analogous: the spatial and spectral subspaces of the input act as competing "modalities," with the spatial module monopolizing the gradient budget. We discuss this connection further in Section 2.

Outline. Section 2 reviews related work. Section 3 establishes notation. Section 4 proves Theorem 13. Section 5 proves Theorem 28. Section 6 formalizes the EGR diagnostic. Section 7 validates both theorems and the diagnostic on synthetic spectral-spatial problems with linear, CNN, and ViT spatial models. Section 8 briefly connects the synthetic findings to real FTIR/QCL tissue classification data. Section 9 discusses prescriptions, limitations, and open problems.

2 Related Work

Our analysis sits at the intersection of four lines of research: the implicit bias and gradient dynamics of cross-entropy training; the geometry of deep-network loss surfaces; multimodal gradient competition; and spectral-spatial deep learning for biomedical and remote-sensing applications. We survey each in turn.

2.1 Gradient starvation and implicit bias

Cross-entropy minimization is known to induce structural biases in the learned representation. Soudry et al. Soudry et al. [2018] proved that on linearly separable data, gradient descent on logistic loss converges to the maximum-margin classifier. Gunasekar et al. Gunasekar et al. [2017] and Lyu and Li Lyu and Li [2020] extended this picture to deep homogeneous networks. These results establish that gradient-based training selects *which* solution among many that fit the training data.

Pezeshki et al. Pezeshki et al. [2021] identified a distinct phenomenon termed *gradient starvation*: in the Neural Tangent Kernel regime, dominant features in the input rapidly saturate the cross-entropy loss, after which weaker but task-relevant features receive negligible gradient signal. Our Theorem 28 adapts this framework to compositional architectures, where the "dominant features" are introduced not by the input distribution but by an architectural capacity asymmetry between modules. The result is a module-level starvation rather than a feature-level one.

2.2 Two-timescale stochastic approximation

The classical machinery for analyzing coupled gradient flows with disparate timescales is Borkar's stochastic approximation framework Borkar [2008]. Heusel et al. Heusel et al. [2017] adapted this to neural network training by introducing the Two Time-Scale Update Rule for GANs, proving convergence to a local Nash equilibrium under explicit step-size disparity. Our setting differs structurally: the timescale separation in our compositional model is not imposed by the optimizer (the same learning rate is applied to both modules) but emerges *implicitly* from the Hessian eigenvalue gap quantified by Theorem 13.

2.3 Hessian geometry of deep networks

The eigenspectrum of the neural-network Hessian has been characterized empirically and theoretically. Sagun et al. Sagun et al. [2017] identified a bulk-plus-outliers structure for overparametrized networks. Pennington and Bahri Pennington and Bahri [2017] developed a random matrix theory analysis that predicts spectral densities under Gaussian assumptions. Karakida et al. Karakida et al. [2019] derived closed-form mean-field expressions for the leading eigenvalue of the Fisher Information Matrix, showing it scales linearly with the layer width. Theorem 13 works blockwise on the joint Gauss–Newton matrix of a compositional model: a width-linear lower bound on the spatial block's top eigenvalue is combined with a width-independent operator-norm cap on the spectral block, yielding a width-scaling law for the module curvature disparity $\mathcal{D}_{\text{curv}}$.

2.4 Multimodal gradient competition

The phenomenon most structurally analogous to ours is gradient competition in multimodal learning. Wang, Tran, and Feiszli Wang et al. [2020] showed empirically that joint training of multimodal classification networks underperforms unimodal training because different modalities overfit and generalize at different rates. They proposed Gradient Blending to dynamically reweight modality-specific losses. Peng et al. Peng et al. [2022] identified "modality laziness" — a tendency for one modality to dominate training, leaving the other under-optimized — and proposed On-the-fly Gradient Modulation. These works are empirical; our contribution adapts and extends their framework to the unimodal *compositional* setting and provides a rigorous geometric foundation. The recent biomedical application of gradient blending to multimodal sarcoma prognosis Bozzo et al. [2024] confirms the mechanism operates in medical settings, but is restricted to explicit dual-modality (MRI + clinical) architectures, not the single-modality compositional pipelines we consider.

2.5 Spectral-spatial deep learning

Spectral-spatial deep learning has been an active area for over a decade, primarily in two communities: biomedical infrared spectroscopy and hyperspectral remote sensing. Berisha et al. Berisha et al. [2019] established the benefit of combining spectral and spatial features for FTIR histology and observed that some tissue types could be classified from spatial features alone, with "minimal chemical information" required — a foreshadowing of the spatial shortcut phenomenon we diagnose. More recently, O'Leary et al. O'Leary et al. [2026] systematically compared spatial-spectral architectures for prostate cancer classification in infrared spectroscopy and reported the striking finding that compressing the spectral dimension to as few as 16 features had negligible effect on all models tested. The authors interpreted this as evidence that tissue classification requires only a limited spectral signature. Our Theorem 28 provides an alternative explanation: end-to-end training fails to extract the spectral signature even when it is present.

The conceptually nearest work in biomedical imaging is the recent characterization of shortcut learning in histopathology by Dawood and Minhas et al. Dawood et al. [2026], who showed that AI models predicting molecular biomarkers from H&E images exploit spurious correlations with clinicopathological features rather than learning genuine biological signal. Our paper extends the shortcut-learning framework into a new modality (spectroscopic imaging) with a different mechanism (gradient competition driven by architectural asymmetry rather than confounder leakage).

2.6 Frozen and random features

Frozen random features have a long history in machine learning. Saxe et al. Saxe et al. [2011] showed that convolutional networks with random (untrained) filters achieve competitive performance on standard benchmarks. Rahimi and Recht Rahimi and Recht [2007] established the theoretical foundations of random features for kernel methods. More recently, Coil and Cheney Coil and Cheney [2025] showed that freezing layers acts as implicit regularization in overparametrized networks. These works establish that frozen random features are not an oddity but a principled choice. Our contribution is to identify the specific mechanism — gradient starvation under capacity asymmetry — under which freezing is not merely competitive but *strictly preferable* to joint training.

3 Setup and Preliminaries

We analyze a generic two-module compositional model. While our motivating application is infrared spectroscopic imaging for tissue classification, the framework applies to any pipeline that first reduces a per-element spectral signal and then applies a high-capacity spatial model. We deliberately keep the notation domain-agnostic.

3.1 The compositional model

Let $X \in \mathbb{R}^{H\times W\times S}$ denote a hyperspectral input image, where H and W are spatial dimensions and S is the spectral dimension.[1] We write the per-pixel spectrum at location (h, w) as $x_{h,w} \in \mathbb{R}^S$.

The model consists of two modules applied sequentially:

1. A *spectral reduction* $f_{\boldsymbol{\theta}} : \mathbb{R}^S \to \mathbb{R}^K$ applied independently to every pixel, producing a feature map $Z \in \mathbb{R}^{K\times H\times W}$ with $Z_{:,h,w} = f_{\boldsymbol{\theta}}(x_{h,w})$.
2. A *spatial model* $g_{\boldsymbol{\phi}} : \mathbb{R}^{K\times H\times W} \to \mathbb{R}^{N_{\text{cls}}\times H\times W}$ producing per-pixel class logits $\hat{y} = g_{\boldsymbol{\phi}}(Z)$.

We denote the composed model $F_{\boldsymbol{\theta},\boldsymbol{\phi}}(X) = g_{\boldsymbol{\phi}}(f_{\boldsymbol{\theta}}(X))$ and the parameter counts $C_f := \dim(\boldsymbol{\theta})$ and $C_g := \dim(\boldsymbol{\phi})$.

Assumption 1 (Linear spectral reduction)**.** $f_{\boldsymbol{\theta}}$ is linear in $\boldsymbol{\theta}$: there exists $W \in \mathbb{R}^{S\times K}$ with $\boldsymbol{\theta} = \text{vec}(W)$ and $f_{\boldsymbol{\theta}}(x_{h,w}) = W^\top x_{h,w}$.

Remark 2. Assumption 1 captures the canonical case used widely in spectroscopy pipelines, including the FTIR and QCL tissue classification work we draw motivation from. It is also the cleanest case in which the Hessian of $\mathcal{L}$ admits closed-form block analysis. We discuss extensions to shallow nonlinear $f_{\boldsymbol{\theta}}$ in Section 9 and verify empirically in Section 7 that the theorem's scaling persists when $f_{\boldsymbol{\theta}}$ is replaced by a small MLP.

Assumption 3 (Spatial model regularity)**.** $g_{\boldsymbol{\phi}}$ is twice continuously differentiable in $\boldsymbol{\phi}$ with L-Lipschitz gradients.

Assumption 4 (Sub-Gaussian inputs)**.** Per-pixel inputs $\{x_{h,w}\}$ are sub-Gaussian with population second moment $\Sigma = \mathbb{E}[x_{h,w}x_{h,w}^\top]$ of bounded condition number. We write

$$\Sigma_X \;:=\; \frac{1}{N}\sum_{n=1}^{N} x_n x_n^\top, \qquad N \;:=\; N_{\text{data}}HW, \tag{1}$$

for the corresponding *mean* empirical Gram over all pixels of all images (x_n enumerating the pixel spectra), so that $\lambda_{\max}(\Sigma_X) \to \lambda_{\max}(\Sigma) < \infty$ as $N \to \infty$. Every constant below is stated in terms of $\lambda_{\max}(\Sigma_X)$ and is therefore independent of the dataset size.

Assumption 5 (Capacity gap)**.** $C_g/C_f \to \infty$.

Assumption 5 is not used in any proof: it is the structural regime in which our theorems become *informative* rather than merely true, and we state it to delimit scope. Empirically it is satisfied with room to spare in modern spectral-spatial architectures: in our motivating example (Section 8, Table 2) the ratio is $C_g/C_f \approx 300$, roughly two orders of magnitude.

Assumption 6 (Bounded input Jacobian of the classifier)**.** There exists a constant $\widetilde{L} > 0$ depending on the depth of $g_{\boldsymbol{\phi}}$, the bottleneck dimension K, the number of classes N_{cls}, and the data distribution — but *not* on the characteristic width M or the total spatial parameter count C_g — such that

$$\sup_{\boldsymbol{\phi}\in\Phi_{\text{reg}},\; Z\in\mathcal{Z}_{\text{reg}}} \left\| \frac{\partial g_{\boldsymbol{\phi}}(Z)}{\partial Z} \right\|_{\text{op}} \;\le\; \widetilde{L},$$

uniformly over the training-time parameter regime Φ_{reg} and the induced feature domain $\mathcal{Z}_{\text{reg}}$.

[1] Multi-channel inputs (e.g. raw absorbance together with first and second derivatives) are handled by concatenating channels along the spectral dimension; this changes only the input size and leaves the analysis unaffected.

Remark 7 (Why Assumption 6 is separate from Assumption 3). Assumption 3 bounds the second derivatives of $g_{\boldsymbol{\phi}}$ with respect to *parameters* $\boldsymbol{\phi}$ (the standard L-smoothness condition of the optimization literature). Assumption 6 bounds the first derivative with respect to *input features* Z—a different object with a different constant. Parameter-side smoothness does not imply input-side Lipschitzness; we require both.

At random Kaiming/Xavier initialization, the operator norm of a variance-scaled random weight matrix is $\Theta(1)$ in width M by the Bai–Yin law Bai and Yin [1993] (see also Vershynin Vershynin [2018], Thm. 4.4.5; Marchenko–Pastur Marčenko and Pastur [1967] describes the corresponding bulk spectrum), giving $\|\partial g_{\boldsymbol{\phi}}/\partial Z\|_{\mathrm{op}} = \Theta(1)$ in width M for standard CNN and MLP blocks. Under the maximal-update parameterization (μP) of Yang and Hu Yang and Hu [2021], the bound persists throughout training as $M \to \infty$. Under standard-parameterization training with weight decay $\lambda > 0$ and bounded-norm iterates, Assumption 6 follows from the boundedness of $\|\boldsymbol{\phi}\|$ combined with the per-layer product-of-operator-norms bound; the constant $\widetilde{L}$ then depends multiplicatively on the network depth (a symbol-free statement: one factor per layer) but not on the width. Softmax self-attention is Lipschitz only on bounded inputs Kim et al. [2021], so $\mathcal{Z}_{\mathrm{reg}}$ should be taken compact whenever $g_{\boldsymbol{\phi}}$ uses attention—this is guaranteed by LayerNorm before attention, which is standard practice.

3.2 Loss and gradients

We train with per-pixel cross-entropy, averaged over all N pixel positions of the dataset:

$$\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi}) = \frac{1}{N} \sum_{n=1}^{N} \mathrm{CE}(\hat{y}_n,\ y_n), \qquad \hat{y}_n = F_{\boldsymbol{\theta},\boldsymbol{\phi}}(X)_n, \tag{2}$$

where $y_n \in \{1, \ldots, N_{\mathrm{cls}}\}$ is the ground-truth class. Cross-entropy plays an essential role in Section 5: its gradient with respect to the logits $\hat{y}$ is the prediction residual $r_n = \mathrm{softmax}(\hat{y}_n) - y_{n,\mathrm{onehot}}$, which vanishes exponentially in the logit margin as predictions saturate—and hence, on separable data, polynomially in training time (Section 5). We discuss other loss functions sharing this vanishing-residual property in Section 9.

Normalization convention. Consistently with (2), we collect the per-pixel quantities into N-normalized stacked objects: the residual vector $r := N^{-1/2}(r_1, \ldots, r_N)$, so that $\|r\|$ is the root-mean-square per-pixel residual, and the *logit Jacobian*

$$J_\bullet \ := \ N^{-1/2}\, \frac{\partial(\hat{y}_1, \ldots, \hat{y}_N)}{\partial \bullet}, \qquad \bullet \in \{\boldsymbol{\theta}, \boldsymbol{\phi}\}, \tag{3}$$

so that $\nabla_\bullet \mathcal{L} = J_\bullet^\top r$ exactly. Two Jacobian-like objects appear in the literature and we fix both here: $J_\bullet$ above (logit Jacobian), and the *residual Jacobian* $J_\bullet^{\mathrm{res}} := \partial r/\partial \bullet = H_\tau J_\bullet$, where $H_\tau := \mathrm{diag}(p) - pp^\top$ is the softmax Jacobian (blockwise over pixels). Since $0 \preceq H_\tau \preceq I$, any operator-norm bound on $J_\bullet$ applies a fortiori to $J_\bullet^{\mathrm{res}}$. The intermediate factor $\partial Z/\partial \boldsymbol{\theta}$ carries the same $N^{-1/2}$ stacking, so that the chain rule composes as $J_{\boldsymbol{\theta}} = (\partial \hat{y}/\partial Z) \cdot (\partial Z/\partial \boldsymbol{\theta})$ with the unnormalized middle factor — which is the object Assumption 6 bounds. All Gram matrices, and Σ_X in (1), use this same $1/N$ normalization, so no constant in this paper depends on the dataset size.

The gradient of $\mathcal{L}$ with respect to the spatial parameters is computed by standard backpropagation. The gradient with respect to the spectral parameters $\boldsymbol{\theta}$ admits the chain-rule decomposition

$$\nabla_{\boldsymbol{\theta}} \mathcal{L} = \underbrace{r}_{\text{residual}} \cdot \underbrace{\frac{\partial \hat{y}}{\partial Z}}_{\text{spatial Jacobian}} \cdot \underbrace{\frac{\partial Z}{\partial \boldsymbol{\theta}}}_{\text{input}} \ = \ J_{\boldsymbol{\theta}}^\top r, \tag{4}$$

with all three factors in the normalization fixed above (the per-pixel loss gradients $\partial\mathcal{L}/\partial\hat{y}_n = r_n/N$ are absorbed into the $N^{-1/2}$-scaled r and $\partial Z/\partial\boldsymbol{\theta}$). The softmax enters through the residual itself; the matrix H_τ appears explicitly only when differentiating r, as in J^{res} above and in the Gauss-Newton blocks below. The three factors play distinct roles in our analysis. The residual is small when predictions are confident. The spatial Jacobian depends on the current state of $\boldsymbol{\phi}$ and thus changes throughout training. The final factor is the input itself and is stationary. The product structure implies that if the residual decays to zero, the gradient to $\boldsymbol{\theta}$ vanishes regardless of the other two factors. This observation underlies Theorem 28.

Remark 8 (How much of this depends on cross-entropy?). The two theorems depend on the loss to different degrees. Theorem 13 is essentially loss-agnostic: the softmax Jacobian H_τ enters its proof only through the upper bound $H_\tau \preceq I$ and through the per-pixel weights $p_{\min}^{(n)}$ that the head hypothesis carries—both contribute constants and no M-dependence. Under squared error the inner matrix is the identity and the argument simplifies, giving the same width-linear disparity. Theorem 28 is more loss-specific: the polynomial residual envelope of Assumption 25 is the characteristic behavior of logistic-type losses on separable data (Soudry et al. Soudry et al. [2018]), whereas squared error on a well-conditioned target admits a faster (exponential) envelope. As noted after Assumption 25, a faster envelope only tightens the displacement bound; the mechanism—a fast module driving the residual down and starving a slow one through the shared factor—is not specific to cross-entropy, though the $\log(1/\delta)$ rate is.

3.3 The block Hessian

The joint Hessian of $\mathcal{L}$ decomposes into blocks aligned with the two parameter groups:

$$H = \nabla^2\mathcal{L}(\boldsymbol{\theta},\boldsymbol{\phi}) = \begin{pmatrix} H_{\boldsymbol{\theta}\boldsymbol{\theta}} & H_{\boldsymbol{\theta}\boldsymbol{\phi}} \\ H_{\boldsymbol{\phi}\boldsymbol{\theta}} & H_{\boldsymbol{\phi}\boldsymbol{\phi}} \end{pmatrix}, \tag{5}$$

with $H_{\boldsymbol{\theta}\boldsymbol{\theta}} \in \mathbb{R}^{C_f\times C_f}$, $H_{\boldsymbol{\phi}\boldsymbol{\phi}} \in \mathbb{R}^{C_g\times C_g}$, and cross blocks of appropriate sizes. Theorem 13 bounds the *ratio* of the two diagonal blocks' top eigenvalues (Definition 11) in terms of the spatial width, by bounding each block separately—not the condition number of H, which is a different and, here, degenerate quantity (Remark 12).

For curvature analysis we work with the generalized Gauss-Newton matrix, which for softmax cross-entropy is

$$G \;=\; J^\top H_\tau J, \qquad J = [\,J_{\boldsymbol{\theta}},\; J_{\boldsymbol{\phi}}\,], \tag{6}$$

with J the (N-normalized) logit Jacobian of (3) and H_τ the softmax Jacobian (Botev et al. Botev et al. [2017]). G is the positive semidefinite part of H that survives at interpolation, and $H = G$ exactly when the residual vanishes. Theorem 13 is stated for the diagonal blocks $G_{\boldsymbol{\theta}\boldsymbol{\theta}} = J_{\boldsymbol{\theta}}^\top H_\tau J_{\boldsymbol{\theta}}$ and $G_{\boldsymbol{\phi}\boldsymbol{\phi}} = J_{\boldsymbol{\phi}}^\top H_\tau J_{\boldsymbol{\phi}}$. The residual-Gram variant $J^{\mathrm{res}\top} J^{\mathrm{res}} = J^\top H_\tau^2 J$ obeys the same upper bounds unconditionally ($H_\tau^2 \preceq I$) and the same lower bound under a correspondingly weighted form of the head hypothesis (supplement Section A.1), so the results transfer across conventions with only the softmax weighting of the constants changing.

3.4 The Effective Gradient Ratio

We introduce a real-time diagnostic that tracks the asymmetry in gradient flow between the two modules.

Definition 9 (Effective Gradient Ratio). At training step t with parameters $(\boldsymbol{\theta}_t, \boldsymbol{\phi}_t)$, the Effective Gradient Ratio is

$$\mathrm{EGR}(t) \;:=\; \frac{\|\nabla_{\boldsymbol{\theta}}\mathcal{L}(\boldsymbol{\theta}_t,\boldsymbol{\phi}_t)\|}{\|\nabla_{\boldsymbol{\phi}}\mathcal{L}(\boldsymbol{\theta}_t,\boldsymbol{\phi}_t)\|}.$$

Remark 10. EGR is a unitless quantity directly computable inside any training loop with negligible overhead. Section 6 bounds the decay of its numerator—the spectral gradient norm—by $B_T C/(1+\mu t)$ under the hypotheses of Theorem 28, characterizes the ratio's trajectory empirically, and proposes a threshold criterion for detecting gradient starvation onset.

3.5 Notational conventions

Throughout, $\|\cdot\|$ denotes the Euclidean (ℓ_2) norm for vectors and the spectral norm for matrices; $\lambda_{\max}(\cdot)$ and $\lambda_{\min}(\cdot)$ denote the largest and smallest eigenvalues of a symmetric matrix; $\kappa(A) = \lambda_{\max}(A)/\lambda_{\min}(A)$ its condition number. We write $A \succeq B$ to mean $A - B$ is positive semidefinite. The symbol $\Omega(\cdot)$ is used in the standard asymptotic sense.

Definition 11 (Module curvature disparity)**.** For the Gauss-Newton matrix G of (6) with diagonal blocks $G_{\boldsymbol{\phi\phi}}, G_{\boldsymbol{\theta\theta}}$ indexed as in (5), the *module curvature disparity* is

$$\mathcal{D}_{\mathrm{curv}}(G) \;:=\; \frac{\lambda_{\max}(G_{\boldsymbol{\phi\phi}})}{\lambda_{\max}(G_{\boldsymbol{\theta\theta}})},$$

the ratio of the spatial block's top eigenvalue to the spectral block's. Unless stated otherwise, $\mathcal{D}_{\mathrm{curv}}$ is evaluated at random initialization $(\boldsymbol{\theta}_0, \boldsymbol{\phi}_0)$, which is where Theorem 13 bounds it and where Experiment 1.1 measures it.

Remark 12 (Why disparity, not condition number)*.* Theorem 13 is a purely geometric statement about how curvature concentrates between the two modules as the spatial network widens. It is not a joint-Hessian condition-number bound: the joint Gauss-Newton G is rank-deficient (parameter directions orthogonal to the classifier's range have exactly zero curvature), so $\kappa(G) = \lambda_{\max}(G)/\lambda_{\min}(G)$ is trivially infinite. The disparity $\mathcal{D}_{\mathrm{curv}}$ is well-defined regardless of rank structure, uses only well-behaved operator-norm quantities, and is exactly the ratio measured in Experiment 1.1. Its growth with spatial capacity is the geometric mechanism that motivates the two-timescale dynamical analysis of Theorem 28, but the dynamical claim requires its own hypotheses and does not follow from $\mathcal{D}_{\mathrm{curv}}$ alone.

4 Theorem 1: Capacity-Induced Module Curvature Disparity

This section establishes our first main result: the module curvature disparity $\mathcal{D}_{\mathrm{curv}}$ (Definition 11) grows with the spatial module's width. The result is the geometric foundation for the dynamical analysis of Section 5: it identifies a capacity-dependent mechanism capable of generating disparate module learning rates. The dynamical starvation claim itself is stated separately in Theorem 28, under its own explicit hypotheses.

4.1 Main result

Theorem 13 (Capacity-Induced Module Curvature Disparity)**.** *Consider the compositional model $F_{\boldsymbol{\theta},\boldsymbol{\phi}} = g_{\boldsymbol{\phi}} \circ f_{\boldsymbol{\theta}}$ with fixed-dimensional linear spectral module $f_{\boldsymbol{\theta}}$ (Assumption 1) and a family of spatial networks $g_{\boldsymbol{\phi}}^{(M)}$ indexed by characteristic width M (channels for a CNN, embedding dimension for a ViT, hidden width for an MLP). Under Assumptions 1, 4, and 6, together with the head hypothesis of Lemma 15 below (a dense classifier head over penultimate features of dimension $\Theta(M)$ that are correlated across inputs at initialization), there exist constants $c_1, c_2 > 0$ depending on the data distribution, the initialization scheme, the bottleneck dimension K, the input spectral dimension S, and the architecture class of $g_{\boldsymbol{\phi}}$ (depth, activation, head dimensions)—but not on M or on the spatial parameter count C_g—such that*

$$\mathcal{D}_{\mathrm{curv}}^{(M)} \;=\; \frac{\lambda_{\max}(G_{\boldsymbol{\phi\phi}}^{(M)})}{\lambda_{\max}(G_{\boldsymbol{\theta\theta}})} \;\geq\; c_1 \cdot M - c_2$$

in expectation over random Kaiming (variance-scaled Gaussian) initialization. Equivalently, for architecture families with $C_g = \Theta(M^2)$,

$$\mathcal{D}_{\mathrm{curv}}^{(M)} \;\geq\; c_1' \cdot \sqrt{C_g/C_f} - c_2.$$

Remark 14 (Interpretation). Theorem 13 states that as the spatial network widens, its top Gauss-Newton eigenvalue grows without bound while the spectral module's top Gauss-Newton eigenvalue remains capped by a C_g-independent constant. Curvature increasingly concentrates in the downstream module. This is a purely geometric fact about the Hessian; the dynamical consequence—that joint training pins $\boldsymbol{\theta}$ near its initialization—is stated separately in Theorem 28, under its own hypotheses.

The proof decomposes the disparity bound into separate bounds on the top eigenvalues of the two diagonal blocks. We state the supporting lemmas before presenting the proof.

4.2 Supporting lemmas

Lemma 15 (Spatial block scaling)**.** *Let M denote the characteristic width of $g_{\boldsymbol{\phi}}$ (channel count for a CNN, embedding dimension for a ViT, hidden layer width for an MLP), let $N_{\mathrm{cls}} \geq 2$, and suppose $g_{\boldsymbol{\phi}}$ ends in a dense classifier head over penultimate features $h_n \in \mathbb{R}^{M_h}$ with $M_h = \Theta(M)$ whose softmax-weighted mean Gram $S_h^p := N^{-1} \sum_n p_{\min}^{(n)} h_n h_n^\top$ (Equation (11)) satisfies*

$$\lambda_{\max}(S_h^p) \;\geq\; q_h M_h \quad \textit{with probability } \geq 1 - \delta_h$$

over the random initialization—where $p_{\min}^{(n)}$ denotes the smallest softmax probability at pixel n—for constants $q_h > 0$ and $\delta_h < 1$ independent of M (the head hypothesis*: the features carry a common direction of $\Theta(M)$ energy across inputs, on pixels whose initial predictions are non-degenerate. It is a cross-input* correlation *condition, not a per-vector norm condition, and is closely related to the quantity κ_2 carrying the width-linear term in Karakida et al.'s law—see the supplement, where a fully proved instance is also given). Then the spatial Gauss-Newton block of* (6) *satisfies*

$$\lambda_{\max}(G_{\boldsymbol{\phi\phi}}) \;=\; \Omega(M) \qquad \textit{in expectation over random Kaiming (variance-scaled Gaussian) initialization.}$$

The logit Gram $J_{\boldsymbol{\phi}}^\top J_{\boldsymbol{\phi}} \succeq G_{\boldsymbol{\phi\phi}}$ inherits the bound unconditionally; the residual Gram $J_{\boldsymbol{\phi}}^{\mathrm{res}\top} J_{\boldsymbol{\phi}}^{\mathrm{res}}$ satisfies it under the analogous hypothesis on the p^2-weighted Gram $S_h^{p^2}$ (supplement). For architecture families with $C_g = \Theta(M^2)$ (fixed depth with at least two consecutive width-proportional layers), equivalently $\lambda_{\max}(G_{\boldsymbol{\phi\phi}}) = \Omega(\sqrt{C_g})$.

Remark 16. Lemma 15 is proved in the supplement (Section A.1) by an explicit witness: the rank-one head-weight perturbation $V = c\,u^\top$, where c is a unit class contrast and u the dominant direction of the feature ensemble. Evaluating the curvature along V gives the deterministic inequality $\lambda_{\max}(G_{\boldsymbol{\phi\phi}}) \geq \lambda_{\max}(S_h^p)$, and the head hypothesis supplies $\lambda_{\max}(S_h^p) = \Omega(M)$. The width-linear law agrees with the mean-field analysis of Karakida et al. Karakida et al. [2019]: their Theorem 4 gives $\lambda_{\max} = \Theta(M)$ with sharp constant for proportional-width fully connected networks under squared loss, and their Theorem 1 shows the mean eigenvalue simultaneously shrinks as $\mathcal{O}(1/M)$—curvature concentrates in a vanishing fraction of directions as networks widen. The scaling variable is width, not parameter count: for architecture families with $C_g = \Theta(M^2)$, the equivalent parameter-count statement is $\lambda_{\max} = \Omega(\sqrt{C_g})$. The head hypothesis covers per-pixel dense heads on CNN and ViT modules as well; the sharpness of the width-linear rate is tested empirically in Section 7.

Lemma 17 (Eigenvalue interlacing for block matrices). *For any symmetric block matrix $M = \begin{pmatrix} A & B \\ B^\top & D \end{pmatrix}$ with principal blocks A and D,*

$$\lambda_{\max}(M) \geq \max\{\lambda_{\max}(A), \lambda_{\max}(D)\}.$$

Remark 18. Lemma 17 is a standard consequence of Cauchy's interlacing theorem applied to principal submatrices (see e.g. Bhatia Bhatia [1997], Corollary III.1.5).

Lemma 19 (Spectral block operator-norm cap). *Under Assumptions 1 (linear per-pixel $f_{\boldsymbol{\theta}}$), 4 (bounded input second moment), and 6 (bounded input Jacobian $\|\partial g_{\boldsymbol{\phi}}/\partial Z\|_{\mathrm{op}} \leq \widetilde{L}$ uniformly in C_g), the spectral logit Jacobian and Gauss-Newton block satisfy*

$$\lambda_{\max}(G_{\boldsymbol{\theta\theta}}) \;\leq\; \|J_{\boldsymbol{\theta}}\|_{\mathrm{op}}^2 \;\leq\; \widetilde{L}^2 \cdot \lambda_{\max}(\Sigma_X) \;=:\; C_{\boldsymbol{\theta}},$$

where $C_{\boldsymbol{\theta}}$ depends only on the classifier's input Lipschitz constant $\widetilde{L}$ and the top eigenvalue of the mean per-pixel input Gram Σ_X of (1)*—not on the spatial width M, the spatial parameter count C_g, or the dataset size N. The same bound holds for the residual-Gram convention, since $H_\tau^2 \preceq H_\tau \preceq I$.*

Remark 20. Lemma 19 follows from the chain-rule identity $J_{\boldsymbol{\theta}} = J_Z \cdot (\partial Z/\partial \boldsymbol{\theta})$ for the logit Jacobian, combined with $\|J_Z\|_{\mathrm{op}} \leq \widetilde{L}$ (Assumption 6) and the exact form of $\partial Z/\partial \boldsymbol{\theta}$ under linearity of $f_{\boldsymbol{\theta}}$ (Assumption 1, giving the per-pixel factor $I_K \otimes x_n^\top$, whose N-normalized stacking has $\|\partial Z/\partial \boldsymbol{\theta}\|_{\mathrm{op}}^2 = \lambda_{\max}(\Sigma_X)$ exactly, by $I_K \otimes \Sigma_X$). Passing from the logit Jacobian to the Gauss-Newton block or the residual Jacobian only inserts factors of $H_\tau \preceq I$ and can only decrease the bound. The bound is on the operator norm (equivalently, the top eigenvalue), which is well-behaved under rank-deficiency of the block—avoiding the smallest-eigenvalue and Kronecker-factorisation machinery discussed in Section A.1.

4.3 Proof of Theorem 13

Proof of Theorem 13. By Lemma 15, in expectation over random Kaiming (variance-scaled Gaussian) initialization,

$$\lambda_{\max}(G_{\boldsymbol{\phi\phi}}^{(M)}) \;\geq\; c_\phi \cdot M$$

for a constant $c_\phi > 0$ independent of M (depending on the initialization scheme, the activation, the head dimensions, and the data distribution; supplement Section A.1). By Lemma 19,

$$\lambda_{\max}(G_{\boldsymbol{\theta\theta}}) \;\leq\; \|J_{\boldsymbol{\theta}}\|_{\mathrm{op}}^2 \;\leq\; C_{\boldsymbol{\theta}}$$

for a constant $C_{\boldsymbol{\theta}} > 0$ independent of M. Dividing,

$$\mathcal{D}_{\mathrm{curv}}^{(M)} \;=\; \frac{\lambda_{\max}(G_{\boldsymbol{\phi\phi}}^{(M)})}{\lambda_{\max}(G_{\boldsymbol{\theta\theta}})} \;\geq\; \frac{c_\phi \cdot M}{C_{\boldsymbol{\theta}}} \;=\; c_1 \cdot M$$

with $c_1 := c_\phi / C_{\boldsymbol{\theta}}$. A constant $c_2 \geq 0$ absorbs finite-width corrections. Under $C_g = \Theta(M^2)$ (families with at least two consecutive width-proportional layers), $M = \Theta(\sqrt{C_g})$ and the ratio form $c_1' \cdot \sqrt{C_g/C_f}$ follows on multiplying and dividing by $\sqrt{C_f}$. This completes the proof. □

4.4 Implications

Theorem 13 establishes that curvature concentrates in the downstream spatial module as its width grows. We highlight two implications, and separate them explicitly from the dynamical consequences of Section 5.

Geometric mechanism, not a dynamical claim. $\mathcal{D}_{\text{curv}} \gg 1$ identifies a *potential* for disparate module learning rates: the spatial Gauss-Newton block admits directions of arbitrarily large curvature while the spectral block does not. Whether joint gradient descent actually produces disparate effective learning rates depends on an additional dynamical hypothesis (that the shortcut pathway fits the residual at rate μ; Assumption 25), which Section 5 introduces and Experiment 1.2 tests. Theorem 13 does not, by itself, prove that $\boldsymbol{\theta}$ is pinned to initialization. In particular, $\mathcal{D}_{\text{curv}}$ concerns the *top* of the spatial block's spectrum, while the fitting rate μ of Section 5 is governed by how fast the residual can be driven down—a property of other parts of the spatial spectrum. The two are linked only under spectral-shape assumptions we do not make; the theorems are deliberately independent, and Experiments 1.1–1.2 test them separately.

The disparity is invariant to shared step-size tuning. The disparity bound depends only on architectural parameters, not on the choice of optimizer or the shared step-size schedule—so no choice of shared learning rate changes the *geometry*; whether any shared-step-size method escapes the *dynamical* consequence is a separate question we treat as a conjecture below. Adaptive methods (Adam, RMSProp) rescale per-coordinate gradient magnitudes but do not alter the underlying eigenvalue disparity between the two blocks. Two-timescale optimizers with explicitly different learning rates per module (Heusel et al. Heusel et al. [2017]; Borkar 2008 Borkar [2008] Ch. 6) are the natural remedy at the optimizer level. We discuss this in Section 9.

4.5 Relation to classical ill-conditioning

Ill-conditioned optimization has classical causes with classical remedies, and it is natural to ask whether they apply here. They do not, because the disparity of Theorem 13 is driven by the *architecture*, not by the data or the parametrization of a single block.

Feature normalization. Scale asymmetry or collinearity among input features degrades the conditioning of Σ_X and is cured by standardizing inputs. Assumption 4 already grants this cure: we assume the population second moment Σ has bounded condition number, i.e. well-conditioned (e.g. normalized) inputs. The disparity persists regardless, because its growth is driven by the width-scaling of the spatial block (Lemma 15), which is untouched by any transformation of the input features.

Weight decay. Adding $\alpha \|w\|^2$ to the loss shifts the Hessian by $2\alpha I$, lifting every eigenvalue by 2α and famously repairing classical condition numbers. But the shift acts on *both* blocks equally: the disparity becomes $(\lambda_{\max}(G_{\boldsymbol{\phi\phi}})+\alpha)/(\lambda_{\max}(G_{\boldsymbol{\theta\theta}})+\alpha)$, which remains $\Theta(M)$ for any fixed α as M grows. Repairing the disparity would require $\alpha \sim \lambda_{\max}(G_{\boldsymbol{\phi\phi}})$, a regularization strength that would dominate the data term and prevent fitting. Consistent with this, Experiment 1.6 finds that Pezeshki's Spectral Decoupling—an L_2 penalty on logits designed to combat saturation—does not reduce the joint-training collapse, while freezing does.

Adaptive optimizers. Adam and RMSProp precondition per-coordinate gradient magnitudes with a diagonal rescaling. A diagonal preconditioner can equalize per-parameter step sizes but does not restructure the eigenvalue disparity between the two blocks, which is a statement about the blocks' spectra rather than their diagonals. Empirically, the canonical dynamics experiment (Experiment 1.2) is run under Adam and the joint-vs-frozen gap persists at every width; we treat the stronger claim—that no shared-learning-rate adaptive method can remove the pathology—as a conjecture rather than a theorem. Optimizers with explicitly different learning rates per module (two-timescale update rules; Heusel et al. Heusel et al. [2017]) are the natural remedy at the optimizer level and correspond, in the limit, to our freezing prescription.

Remark 21 (Parameterization scope). $\mathcal{D}_{\mathrm{curv}}$ is not invariant to per-module reparameterization, and Theorem 13 is a statement about the standard variance-scaled (Kaiming) parameterization—the one used in the pipelines we analyze and in all our experiments. Under alternative parameterizations that rescale layers with width, such as μP Yang and Hu [2021], both the initialization magnitudes and the effective per-layer learning rates change, and the width scaling of the blocks would need to be re-derived; we do not claim the disparity law transfers. Our two uses of μP elsewhere are limited to a different role: as one route to keeping the input-Jacobian bound (Assumption 6) uniform in width during training. Whether principled reparameterization can neutralize the disparity—rather than merely re-expressing it—is an open question adjacent to the per-module learning-rate remedies discussed in Section 9.

Remark 22 (Small eigenvalues: irrelevance vs. starvation). A parameter direction can carry vanishing curvature for two very different reasons: (a) the direction genuinely does not affect predictions on the task at hand ("irrelevance"), in which case zero gradient is the correct outcome and the parameter should not be learned; or (b) the direction *would* affect predictions if the optimizer could reach it, but the compositional geometry buries the gradient signal behind the bottleneck ("starvation"). The Gauss-Newton bounds of Theorem 13 alone cannot distinguish between these interpretations: both produce C_g-independent small curvature in the spectral block.

The synthetic experiments in Section 7 are designed to distinguish these hypotheses by construction. In Experiment 1.3 we generate data where each of the two modalities alone carries information sufficient to solve the task (single-modality baselines reach ≈ 0.53 on a binary classification, above the 0.50 chance floor). Joint training on the same data does not merely under-use the spectral channel: it undershoots the weaker of the two single-modality baselines. The gap is modest in absolute terms on this synthetic problem—an important honest caveat—but its direction is inconsistent with interpretation (a), under which joint training would at worst match the stronger single-modality baseline.

For real spectroscopic data, distinguishing (a) from (b) requires demonstrating that a frozen pretrained spectral encoder outperforms one that has been jointly trained end-to-end on the same task. We report such experiments on FTIR and QCL tissue classification in Section 8. Our present study compares only two concrete mitigation strategies (freezing the spectral module, and Pezeshki's Spectral Decoupling Pezeshki et al. [2021]); broader comparisons against modality-specific learning rates, gradient surgery, GradNorm-style loss balancing, and related interventions are important future work.

4.6 Empirical support

Experiment 1.1 in Section 7 measures $\mathcal{D}_{\mathrm{curv}}^{(M)}$ directly across a range of spatial widths and confirms monotonic growth with M: the spatial top eigenvalue grows $74\times$ over the sweep while the spectral top eigenvalue stays flat, driving the disparity into the thousands. The measured log–log slope of λ_ϕ against C_g is 0.70 ± 0.05, below the slope-1.0 prediction that applies to the tested architecture family (whose $C_g = \Theta(M)$); the deficit and its candidate causes are reported in Section 7. The qualitative prediction—unbounded spatial-block growth against a capped spectral block—is unambiguous in the data.

Remark 23 (Defense against linear triviality). A natural objection is that, since $f_{\boldsymbol{\theta}}$ is linear, the analysis must reduce to principal component analysis (PCA) of the input. This is not the case. PCA finds directions of maximum input variance *without reference to labels.* Theorem 13 concerns the curvature of the supervised cross-entropy loss, which depends on the current state of $g_{\boldsymbol{\phi}}$ through the Jacobian $\partial\hat{y}/\partial Z$. The pathology we identify is precisely that this supervised signal fails to reach $\boldsymbol{\theta}$, not that $f_{\boldsymbol{\theta}}$ is incapable of representing useful features.

5 Theorem 2: Finite-Time Spectral Starvation

Theorem 13 is a geometric statement: as the spatial module widens, curvature concentrates in it and the spectral block's top curvature stays capped. This section states the dynamical consequence we actually observe in training—the spectral module barely moves during the period in which the training loss is being fit—as a theorem with *explicit dynamical hypotheses*. We deliberately do not claim that the geometry of Theorem 13 alone implies the dynamics: the bridge from curvature disparity to rate separation is stated as a hypothesis, motivated by Theorem 13 and verified empirically in Section 7. Deriving it from the architecture is left as an open problem (Section 9).

Two classical facts shape the formulation. First, for unregularized cross-entropy on separable data there is *no finite minimizer*: the loss decays like $1/t$ while the parameters diverge along the max-margin direction (Soudry et al. Soudry et al. [2018]). A theorem asserting convergence to a finite stationary point would therefore be vacuous or false in the regime modern networks inhabit. Second, the starvation phenomenon of Pezeshki et al. Pezeshki et al. [2021] is fundamentally about which features receive learning signal *during training*, not about the $t \to \infty$ limit. Both point the same way: the honest statement is a *finite-time* bound on how far the spectral parameters can move before the shortcut pathway has fitted the labels.

5.1 Setting and hypotheses

We analyze the continuous-time limit of joint training with a shared learning rate, in which the parameters $(\boldsymbol{\theta}_t, \boldsymbol{\phi}_t)$ evolve by gradient flow

$$\dot{\boldsymbol{\theta}}(t) = -\nabla_{\boldsymbol{\theta}}\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi}), \qquad \dot{\boldsymbol{\phi}}(t) = -\nabla_{\boldsymbol{\phi}}\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi}). \tag{7}$$

The discrete-SGD extension (Robbins–Monro tracking with noise entering the constants) is standard (Borkar Borkar [2008], Ch. 2) and discussed in the supplement.

Recall from the chain rule (4) that $\nabla_{\boldsymbol{\theta}}\mathcal{L} = J_{\boldsymbol{\theta}}^\top r$ where r is the softmax residual, so

$$\left\|\dot{\boldsymbol{\theta}}(t)\right\| \;=\; \left\|J_{\boldsymbol{\theta}}(t)^\top r(t)\right\| \;\le\; \|J_{\boldsymbol{\theta}}(t)\|_{\mathrm{op}} \cdot \|r(t)\|\,. \tag{8}$$

Fix a horizon $T > 0$ and suppose the trajectory remains in the regime of Assumption 6, i.e. $\boldsymbol{\phi}(t) \in \Phi_{\mathrm{reg}}$ and $Z(t) \in \mathcal{Z}_{\mathrm{reg}}$ for all $t \le T$ (*trajectory containment*). Under Assumptions 1, 4, and 6, Lemma 19 then caps the first factor over that horizon:

$$B_T \;:=\; \sup_{t\le T} \|J_{\boldsymbol{\theta}}(t)\|_{\mathrm{op}} \;\le\; \sqrt{C_{\boldsymbol{\theta}}} \;=\; \widetilde{L}\,\sqrt{\lambda_{\max}(\Sigma_X)}, \tag{9}$$

a constant independent of the spatial width M, the parameter count C_g, and the horizon T. (Even without containment, $B_T < \infty$ for every finite T by continuity of $J_{\boldsymbol{\theta}}$ along the flow; containment is what makes the cap *M-independent*, which is the load-bearing property.) We write B for B_T when the horizon is clear. The remaining hypothesis concerns the second factor.

Remark 24 (Containment and the unregularized separable regime). Trajectory containment is a genuine restriction. In the unregularized separable regime that motivates the polynomial envelope below, $\|\boldsymbol{\phi}(t)\| \to \infty$ logarithmically (Soudry et al. Soudry et al. [2018]), so a bound on $\|\partial g_{\boldsymbol{\phi}}/\partial Z\|_{\mathrm{op}}$ that is uniform over *all* t cannot be assumed. Two standard settings restore it: weight decay or norm constraints (bounded iterates; strictly, these modify the flow (7), and the theorem then applies to the modified flow, whose residual envelope may differ), and the maximal-update parameterization Yang and Hu [2021], which leaves the flow unchanged (bound uniform in width). In the pure max-margin limit, $\widetilde{L}$ grows at most logarithmically in T, which turns the displacement bound of Theorem 28 into $\mathcal{O}(\varepsilon \log^2(1/\delta))$—the same conclusion up to a log factor.

Assumption 25 (Shortcut residual decay)**.** There exist constants $C \geq 1$ and $\mu > 0$ such that, along the joint trajectory (7),

$$\|r(t)\| \;\leq\; \frac{C}{1+\mu t},$$

where $\|r(t)\|$ is the root-mean-square per-pixel residual of (3). We refer to μ as the *shortcut fitting rate*. The constants C and μ may depend on the initialization, the data, and the architecture—in particular μ is expected to grow with spatial capacity (Remark 26)—and are not assumed uniform over any family; the theorem holds for any trajectory admitting such an envelope.

Remark 26 (Why polynomial, and where μ comes from)*.* The $1/t$ envelope is the correct cross-entropy-compatible form: for logistic-type losses on separable data the training loss—and hence the residual norm—decays polynomially, not exponentially (Soudry et al. Soudry et al. [2018]). An exponential envelope, where valid (e.g. regularized or non-separable regimes), only strengthens the conclusion below. The hypothesis encodes the *shortcut* mechanism: the residual is driven down by the spatial module exploiting whatever signal is present in the feature map Z (positional and contextual structure), without requiring $f_{\boldsymbol{\theta}}$ to specialize—the spatial-signal sufficiency discussed in Section 5.4. The rate μ is expected to *grow* with spatial capacity—wider networks fit faster, the familiar regularity of over-parameterized training; kernel-regime analyses tie fitting rates to spectral properties of the model's kernel that improve with width (cf. the starvation analysis of Pezeshki et al. Pezeshki et al. [2021]), though we use no such derivation here. We test the capacity dependence of the fitting speed empirically in Experiment 1.2. A derivation of $\mu(C_g)$ for the linearized fast subsystem is given informal treatment in the supplement and is not needed for the theorem, which takes μ as given.

Definition 27 (Fitting time)**.** For a target residual level $\delta \in (0, C)$, the fitting time is

$$T_\delta \;:=\; \inf\{\, t \geq 0 : \|r(t)\| \leq \delta \,\}.$$

Under Assumption 25, $T_\delta \leq (C/\delta - 1)/\mu$.

5.2 Main result

Theorem 28 (Finite-Time Spectral Starvation)**.** *Fix a horizon $T \geq 0$ and suppose the gradient flow* (7) *is contained in $\Phi_{\mathrm{reg}} \times \mathcal{Z}_{\mathrm{reg}}$ up to time T. Under Assumptions 1, 4, 6, and 25, with B_T as in* (9)*, the flow satisfies*

$$\|\boldsymbol{\theta}(T) - \boldsymbol{\theta}_0\| \;\leq\; \frac{B_T\, C}{\mu}\, \log(1 + \mu T),$$

and, whenever $T \geq T_\delta$, at the fitting time T_δ,

$$\|\boldsymbol{\theta}(T_\delta) - \boldsymbol{\theta}_0\| \;\leq\; \frac{B_T\, C}{\mu}\, \log\!\left(\frac{C}{\delta}\right).$$

In particular, writing $\varepsilon := B_T/\mu$ for the effective module-rate ratio, the spectral displacement by the time the training residual reaches level δ is $\mathcal{O}(\varepsilon \log(1/\delta))$.

Proof. By (8), the operator-norm cap (9), and Assumption 25, for every $t \leq T$,

$$\left\|\dot{\boldsymbol{\theta}}(t)\right\| \;\leq\; B_T \cdot \|r(t)\| \;\leq\; \frac{B_T\, C}{1 + \mu t}.$$

Integrating,

$$\|\boldsymbol{\theta}(T) - \boldsymbol{\theta}_0\| \;\leq\; \int_0^T \left\|\dot{\boldsymbol{\theta}}(t)\right\| dt \;\leq\; B_T\, C \int_0^T \frac{dt}{1+\mu t} \;=\; \frac{B_T\, C}{\mu} \log(1 + \mu T).$$

For the second claim, Definition 27 gives $1 + \mu T_\delta \leq C/\delta$, so $\log(1 + \mu T_\delta) \leq \log(C/\delta)$. ☐

5.3 Interpretation

Theorem 28 is a statement about *training-time* behavior, which is what practice cares about: training runs end when the loss is fit (or by early stopping), not at $t = \infty$.

The starvation regime. The displacement bound is a product of factors with opposite capacity dependence. The factor B_T is capped independently of C_g (Lemma 19): no matter how large the spatial module, the spectral gradient cannot exceed $B_T \|r\|$. The fitting rate μ is *expected* to grow with spatial capacity (Remark 26; this is an empirical regularity we test, not a theorem): wider shortcut pathways drive the residual down faster. For families of architectures along which the envelope amplitude C stays bounded—as it does empirically, since $\|r(0)\|$ is $\Theta(1)$ at near-uniform initial predictions and the observed residuals decay from the start rather than plateauing—the displacement at the fitting time scales as $\varepsilon \log(1/\delta)$ with $\varepsilon = B_T/\mu \to 0$, and the spectral module is increasingly pinned to its initialization *for the entire useful duration of training.* We state the boundedness of C along the family as an explicit condition rather than a consequence: Assumption 25 deliberately does not make its constants uniform over architectures, so the capacity-asymptotic reading requires it. This is the capacity-worsens-starvation prediction tested in Experiment 1.2.

What we do not claim. We make no claim about the $t \to \infty$ limit. For unregularized cross-entropy on separable data no finite stationary point exists (Soudry et al. Soudry et al. [2018]), and $\boldsymbol{\theta}$ may eventually move substantially over horizons exponentially longer than any practical training run ($T = \mathcal{O}(\varepsilon^{-q})$ still yields displacement $\mathcal{O}(\varepsilon \log(1/\varepsilon)) \to 0$, but larger horizons are unconstrained). The finite-time framing is not a weakness: it is the phenomenon. A spectral module that receives its learning signal only after the loss has been fitted receives it too late to shape which features the classifier uses.

The logarithm is benign. For any polynomial training horizon $T = \mathcal{O}(\varepsilon^{-q})$ with fixed q, and along families with bounded envelope amplitude C as above, the displacement is $\mathcal{O}(\varepsilon \log(1/\varepsilon))$, which vanishes as $\varepsilon \to 0$. The bound is thus meaningful across every practically reachable horizon, not merely at a single stopping time.

Relation to lazy training. Structurally, Theorem 28 is a *module-wise lazy training* bound: a bounded module Jacobian multiplied by an integrable residual envelope yields logarithmic parameter movement, in the spirit of Chizat, Oyallon, and Bach Chizat et al. [2019]. The differences are the driver and the scope. In the lazy-training literature, small movement follows from a scaling of the *model* (width or output scale) that makes the linearization accurate for *all* parameters; here, movement is small for *one module only*, it is driven by the interaction of two module-asymmetric quantities—the operator-norm cap of Lemma 19 and the shortcut fitting rate—and the other module is entirely free to travel. The phenomenon is a *selective* laziness induced by the compositional position of the spectral module, not a global linearization regime.

5.4 On the hypotheses

The operator-norm cap is proven, not assumed. The factor B is supplied by Lemma 19 under Assumptions 1 and 6; it is the point where Theorem 13's geometry genuinely enters the dynamics.

Residual decay encodes the shortcut. Assumption 25 is a statement about the *residual only*: the training loss gets fitted at rate μ along the joint trajectory. It does not mention $\boldsymbol{\theta}$, and in particular it does not assume the spectral module stays put. The shortcut interpretation—that the fitting is done by the spatial module exploiting structure already present in Z (positional cues, contextual patterns)—is the mechanism that makes the assumption plausible at high spatial capacity, and it is satisfied in practice whenever the spatial model can overfit the training set, the norm for modern over-parameterized CNNs and ViTs. We test the envelope itself empirically in Experiment 1.2.

Remark 29 (Is the hypothesis circular?). A natural worry is that assuming fast residual decay comes close to assuming the conclusion. It does not, and the direction of the implication is the theorem's content. The assumption constrains what the *loss* does; the theorem then caps how far $\boldsymbol{\theta}$ can have travelled *on any trajectory consistent with that loss behavior*—including trajectories on which $\boldsymbol{\theta}$ itself does part of the fitting. Nothing in the hypothesis privileges the spatial module: if a trajectory fits the residual at rate μ partly through spectral learning, the bound still applies and shows the total spectral displacement incurred in doing so is at most $(B_T C/\mu)\log(C/\delta)$. Read contrapositively, this is the sharpest form of the claim: *substantial spectral learning requires slow fitting.* A trajectory on which $\boldsymbol{\theta}$ moves far must have small μ—the loss must stay unfit for a long time—which is precisely what freezing-based pipelines engineer and what fast-fitting joint training precludes. The hypothesis is also independently checkable: the envelope is a measurable property of a training run, verified or refuted by the run's own residual curve, with no reference to $\boldsymbol{\theta}$.

Rate separation is motivated, not derived. Theorem 13 shows curvature disparity grows with capacity; it does not by itself prove that μ grows or that the trajectory satisfies Assumption 25. We keep the two theorems logically independent: Theorem 13 identifies a capacity-dependent geometric mechanism *capable* of generating disparate module learning rates; Theorem 28 characterizes the starvation dynamics *when* such a separation is present. Experiments 1.1–1.3 test the mechanism and the dynamics separately. Deriving Assumption 25 from the architecture (e.g. via an NTK analysis of the fast subsystem at large width, adapting Pezeshki et al. Pezeshki et al. [2021]) is a natural next step; a sketch is given in the supplement.

[TODO: Supplement: (i) discrete-SGD extension of Theorem 28 via Robbins–Monro tracking (Borkar Ch. 2), noise entering the constants; (ii) informal NTK derivation of $\mu(C_g)$ for the linearized fast subsystem; (iii) empirical verification that $\|r(t)\|$ on our synthetic trajectories is enveloped by $C/(1+\mu t)$.]

6 The Effective Gradient Ratio Diagnostic

Theorems 13 and 28 together identify a training pathology that is invisible to *single-run* loss-curve monitoring. The failure is counterfactual: a jointly trained model converges to an apparently healthy solution—training loss down, validation metrics respectable, every curve shaped like a normal run—that is nonetheless worse than what the *same architecture* achieves with its spectral module frozen. Nothing in the run's own curves flags this, because the deficit is visible only against a baseline the practitioner has not trained. One discovers it by training the frozen counterpart (twice the compute, and only if one thinks to try), or not at all.

This section introduces the *Effective Gradient Ratio* (EGR), a real-time diagnostic that detects gradient starvation as it occurs. EGR is a single scalar, computable inside any training loop at negligible cost, that converts the finite-time starvation prediction of Theorem 28 into a concrete observable.

6.1 Definition and basic properties

Recall from Definition 9 that at training step t,

$$\mathrm{EGR}(t) = \frac{\|\nabla_{\boldsymbol{\theta}}\mathcal{L}(\boldsymbol{\theta}_t, \boldsymbol{\phi}_t)\|}{\|\nabla_{\boldsymbol{\phi}}\mathcal{L}(\boldsymbol{\theta}_t, \boldsymbol{\phi}_t)\|}.$$

The numerator and denominator are computable directly from gradients accumulated during the standard backward pass. EGR is unitless and invariant to the choice of learning rate.

Initial value: a heuristic. A useful rule of thumb for the starting point of the diagnostic is

$$\mathrm{EGR}(0) \;\sim\; \sqrt{C_f/C_g} \qquad \text{(heuristic, not a theorem).} \tag{10}$$

The reasoning is a parameter-counting one: at variance-scaled initialization each module's gradient has entries of comparable typical magnitude, so the squared gradient norms scale with the number of parameters carrying them, $\|\nabla_{\boldsymbol{\phi}}\mathcal{L}\|^2 \sim C_g$ against $\|\nabla_{\boldsymbol{\theta}}\mathcal{L}\|^2 \sim C_f$, and the square root comes from passing to norms. We state (10) as a heuristic rather than a proposition for three reasons: it exchanges the expectation of a ratio for the ratio of expectations without a concentration argument; the per-entry-magnitude premise is parametrization-dependent and would change under μP, which we cite elsewhere Yang and Hu [2021]; and the spectral module's gradient additionally passes through the bottleneck, which the count ignores. Its role in this paper is limited to setting the scale for the *relative* threshold of Definition 33, which compares $\mathrm{EGR}(t)$ against the measured $\mathrm{EGR}(0)$ of the run at hand and so does not depend on (10) being sharp. The corresponding numerical value for our motivating architecture follows from the capacity ratio reported in Section 8.

6.2 Decay of the spectral gradient

Proposition 30 (Spectral gradient decay)**.** *Under the hypotheses of Theorem 28—Assumptions 1, 4, 6, and 25, with the trajectory contained in $\Phi_{\mathrm{reg}} \times \mathcal{Z}_{\mathrm{reg}}$ up to a horizon T—we have, for every $t \le T$,*

$$\|\nabla_{\boldsymbol{\theta}}\mathcal{L}(t)\| \;\le\; \frac{B_T\, C}{1+\mu t}, \qquad B_T \;\le\; \sqrt{C_{\boldsymbol{\theta}}} \;=\; \widetilde{L}\sqrt{\lambda_{\max}(\Sigma_X)},$$

with B_T as in (9), $C_{\boldsymbol{\theta}}$ the cap of Lemma 19, and C, μ the constants of Assumption 25.

Proof. Identical to (8): $\|\nabla_{\boldsymbol{\theta}}\mathcal{L}\| = \left\|J_{\boldsymbol{\theta}}^\top r\right\| \le \|J_{\boldsymbol{\theta}}\|_{\mathrm{op}}\, \|r\| \le B_T\, \|r\|$ for $t \le T$, and Assumption 25 bounds $\|r(t)\|$. ☐

Remark 31. The horizon restriction is not a technicality inherited for tidiness: Remark 24 notes that in the unregularized separable regime—the regime Assumption 25 models—a cap uniform over all $t \in [0, \infty)$ cannot be assumed. In practice T is the training run's length, which is exactly the horizon the diagnostic operates over.

Remark 32 (From gradient decay to the EGR ratio)*.* Proposition 30 bounds the *numerator* of the EGR. The denominator $\|\nabla_{\boldsymbol{\phi}}\mathcal{L}\| = \left\|J_{\boldsymbol{\phi}}^\top r\right\|$ is also proportional to the residual, so both quantities vanish together as the loss is fitted, and the behavior of their ratio is not determined by residual decay alone: it depends on the relative conditioning of the two Jacobians along the trajectory. We therefore treat the EGR trajectory as an empirical object, characterized in Section 6.5, and use Proposition 30 only for the absolute claim that the spectral module's learning signal decays polynomially while the loss is being fitted.

6.3 Threshold criterion

The absolute decay of the spectral gradient (Proposition 30), combined with the empirical behavior of the ratio characterized in Section 6.5, motivates a practical criterion for detecting the onset of gradient starvation. We emphasize its status: the threshold rule is an empirically calibrated heuristic that the theory motivates but does not derive (Remark 32).

Definition 33 (Gradient starvation alert)**.** We say the model is in a *gradient starvation regime* at step t if

$$\mathrm{EGR}(t) \quad < \quad \tau \cdot \mathrm{EGR}(0),$$

where $\tau \in (0, 1)$ is a user-supplied tolerance. Empirically, we recommend $\tau \in [0.05, 0.1]$, corresponding to a 10–20× drop in EGR relative to initialization.

Remark 34. The threshold τ is the only hyperparameter of the diagnostic. The default $\tau = 0.1$ corresponds to a one-decade decay and is the value we use throughout Section 7.

Practical use. A practitioner logs $\mathrm{EGR}(t)$ at each step. When the alert fires, two options are available:

1. **Freeze the spectral module.** This is the prescription most directly motivated by Theorem 28: removing $\boldsymbol{\theta}$ from the optimization eliminates the timescale gap. We discuss this in Section 9.

2. **Inject a fixed residual baseline.** A softer intervention: replace $Z = f_{\boldsymbol{\theta}}(X)$ with $Z = f_{\boldsymbol{\theta}}(X) + h(X)$ for a fixed feature extractor h (e.g. PCA). The fixed baseline ensures the spatial model receives stable input even if $f_{\boldsymbol{\theta}}$ continues to drift, breaking the shortcut without freezing.

We evaluate both interventions in Experiment 1.3.

6.4 Relationship to existing diagnostics

EGR is conceptually related to several existing notions but is distinct in its scope and intent.

GradNorm Chen et al. [2018]. GradNorm dynamically reweights losses in multi-task learning to equalize gradient magnitudes across tasks. EGR measures rather than intervenes, and applies to single-task compositional architectures rather than multi-task losses.

Modality-specific learning rates (MSLR). In multimodal learning, MSLR adjusts learning rates per modality. EGR diagnoses the underlying pathology that MSLR addresses but in a unimodal, two-stage architecture where the "modalities" are the spectral and spatial subspaces of a single input.

Loss curve monitoring. The conventional practice of monitoring training and validation loss fails to detect gradient starvation because both can decay even as $\boldsymbol{\theta}$ stagnates. EGR is sensitive to the structural failure even when no global loss curve abnormality is present.

6.5 Empirical validation: EGR as a diagnostic

We validate EGR as a diagnostic of training pathology in two stages. First, we establish that EGR statistics correlate with generalisation across our full sweep. Second, we test whether the signal is genuine or merely a re-encoding of model capacity.

Pooled correlation (capacity confounded). Across $N = 24$ runs spanning four widths $\{16, 64, 256, 1024\}$ with six seeds each, EGR depth correlates strongly with final test accuracy: Pearson $r = -0.723$ ($p = 6.6 \times 10^{-5}$), Spearman $\rho = -0.808$ ($p = 1.8 \times 10^{-6}$), and with the train–test gap at $r = +0.741$ ($p = 3.5 \times 10^{-5}$). However, EGR depth is itself strongly correlated with $\log$ width ($r = +0.903$): roughly 77% of the pooled correlation with test accuracy is attributable to capacity rather than to EGR depth per se (partial r controlling for $\log$ width collapses to -0.169, $p = 0.44$). The pooled correlation alone cannot establish a capacity-independent diagnostic.

Fixed-capacity test. To isolate the capacity-independent signal, we ran a second experiment holding CNN width fixed at $D = 256$ and varying data noise $\{0.05, 0.10, 0.15, 0.20\}$, learning rate $\{3\times10^{-4}, 10^{-3}, 3\times10^{-3}\}$, weight decay $\{0, 10^{-3}\}$, and six seeds, for a total of $N = 144$ runs. At fixed capacity, $\mathrm{EGR}_{\min}$ is the relevant statistic (with width fixed, egr_depth reduces to a constant offset of egr_min in expectation). It exhibits a within-capacity correlation with generalisation that is statistically robust but practically moderate: Pearson $r(\mathrm{EGR}_{\min}, \mathrm{acc}) = +0.344$ ($p = 2.5 \times 10^{-5}$).

The relationship survives every adversarial control we applied. $\mathrm{EGR}_{\min}$ is essentially orthogonal to the swept hyperparameters — marginal correlations with noise, learning rate, and weight decay are $+0.029$, -0.073, and $+0.007$ respectively, with the full hyperparameter set explaining only $R^2 = 0.015$ of $\mathrm{EGR}_{\min}$ variance. The partial correlation $r(\mathrm{EGR}_{\min}, \mathrm{acc} \mid \mathrm{noise}, \mathrm{lr}, \mathrm{wd}) = +0.395$ ($p = 9.5 \times 10^{-7}$) is *larger* than the pooled $r = +0.334$, which is the opposite signature of a hidden hyperparameter confound. The within-bin analysis is the strongest evidence: across the 24 $(\mathrm{noise}, \mathrm{lr}, \mathrm{wd})$ cells in which only the seed varies, 23 of 24 yield strongly positive $r(\mathrm{EGR}_{\min}, \mathrm{acc})$ (mean $+0.584$, Fisher-z aggregated $t = 14.65$, $p < 10^{-13}$).

Controlling for final training loss — itself a non-trivial predictor with $r = +0.47$ for accuracy and $r = -0.55$ for the overfit gap — leaves the $\mathrm{EGR}_{\min}$ partial correlation essentially unchanged at $+0.335$ ($p = 4.1 \times 10^{-5}$). EGR therefore carries information that train-loss monitoring does not.

Practical interpretation. Honesty about effect size is required. Treated as a binary alarm, $\mathrm{EGR}_{\min}$ achieves ROC-AUC $= 0.612$ for flagging below-median test accuracy, and $\mathrm{EGR}_{\mathrm{depth}}$ achieves only 0.538 for high overfit gap. The best operating point ($\mathrm{EGR}_{\min} < 0.10$) catches 90% of bad runs but also flags 57% of healthy runs — no threshold delivers a usable trade-off as a standalone indicator.

Verdict. At fixed capacity, EGR is a real, capacity-independent signal correlated with generalisation. The within-capacity effect size is small enough that EGR is best deployed alongside train-loss monitoring as a complementary diagnostic, not as a standalone run-quality indicator. For research and ablation work, EGR provides genuine insight into the gradient starvation mechanism beyond what loss monitoring reveals; for production monitoring of single training runs, plain loss curves remain the primary signal and EGR a useful adjunct.

6.6 Implementation

The EGR diagnostic is implemented as a PyTorch training callback. It hooks into the standard training loop between the backward pass and the optimizer step:

```
loss.backward()
egr_logger.log_step(global_step)    # <-- the only added line
optimizer.step()
optimizer.zero_grad()
```

The callback identifies the spectral and spatial parameter groups via `model.spectral.parameters()` and `model.spatial.parameters()` (or any user-supplied naming convention) and records the gradient norms and EGR at each step. We release the implementation as the `egr` package, available at the project repository. [TODO: Phase 4, W12: prepare PyPI package skeleton. Add unit tests for gradient norm computation and threshold detection.]

7 Synthetic Experiments

We validate Theorems 13 and 28 and the EGR diagnostic on a controlled synthetic spectral-spatial classification problem. Three experiments isolate distinct theoretical predictions: (1.1) the Hessian condition number bound; (1.2) the two-timescale dynamics under joint training; (1.3) the equal-information killer test demonstrating that joint training prefers spatial features even when spectral and spatial signals carry equal information.

7.1 Synthetic data and architectures

We use a calibrated generator of synthetic hyperspectral images $X \in \mathbb{R}^{H\times W\times S}$ with $H = W = 16$ and $S = 64$, supporting two-class per-pixel classification with controlled spectral and spatial signal strengths through scalar parameters α and β. The data generator places a known spectral direction $u \in \mathbb{R}^S$ in each pixel scaled by a per-pixel latent $z \sim \mathcal{N}(0, 1)$ and modulated by α, with smooth per-class spatial templates modulated by β entering the label-generating logits. Bayes-optimal accuracy is ≈ 0.85–0.95 at $\alpha = \beta = 1$ and noise level 0.1.

For the spatial module $g_{\boldsymbol{\phi}}$ we use three architectures: a two-layer convolutional network (SpatialCNN, 3×3 kernels), a two-layer pre-LN Vision Transformer (SpatialViT, pixel-token), and a per-pixel MLP baseline (SpatialMLP, no spatial receptive field). The MLP is included for ablation; it cannot exhibit a spatial shortcut by construction.

7.2 Experiment 1.1: Hessian capacity ablation

We vary the spatial width D of a SpatialMLP over nearly three orders of magnitude ($D \in \{16, 32, 64, \ldots, 8192\}$, a $512\times$ range) with $C_f = 1024$ held fixed; all measurements in this experiment use the SpatialMLP family. For each (D, seed) we compute the top eigenvalue of the spectral and spatial diagonal blocks of the loss Hessian via power iteration (5 seeds per width). Theorem 13 is stated for the Gauss–Newton blocks; the full-Hessian blocks additionally contain a residual-weighted functional term, and we flag this measurement gap explicitly. The predictions to be tested are $\lambda_{\max}(J_{\boldsymbol{\phi}}^\top J_{\boldsymbol{\phi}}) = \Omega(M)$ (Lemma 15) and $\lambda_{\max}(J_{\boldsymbol{\theta}}^\top J_{\boldsymbol{\theta}})$ approximately constant (Lemma 19).

Findings. Across 50 measurements:

- $\lambda_{\max}(J_{\boldsymbol{\theta}}^\top J_{\boldsymbol{\theta}})$ is flat at ≈ 0.02 over the full range of D – the spectral block does not grow with the spatial capacity, consistent with Lemma 19.
- $\lambda_{\max}(J_{\boldsymbol{\phi}}^\top J_{\boldsymbol{\phi}})$ grows by $74\times$ (from 0.62 to 46) as C_g grows by $510\times$.
- The empirical log-log slope of $\lambda_{\max}(J_{\boldsymbol{\phi}}^\top J_{\boldsymbol{\phi}})$ vs. C_g is 0.70 ± 0.05 across the full range. For the SpatialMLP family $C_g = \Theta(D)$ (input and output dimensions are fixed), so the width-linear law of Lemma 15 predicts slope 1.0 against C_g; the measured 0.70 is sublinear, short of the width-linear asymptote. Candidate causes are finite-width corrections, the loss-Hessian versus Gauss–Newton measurement gap flagged above, and the correlated non-Gaussian features Z relative to the mean-field idealization. The $\sqrt{C_g}$ (slope-0.5) form of Theorem 13 applies only to fixed-depth $\Theta(M^2)$ families, which this sweep does not include.
- The module curvature disparity $\mathcal{D}_{\mathrm{curv}} = \lambda_\phi/\lambda_\theta$ grows from 40 at $C_g/C_f = 0.3$ to 2,407 at $C_g/C_f = 152$. At the capacity ratios of real spectroscopy pipelines (Section 8), this extrapolates to disparities in the thousands.

The qualitative geometric prediction is verified: the spectral and spatial block eigenvalues diverge with capacity, producing a severe module curvature disparity. The measured scaling exponent is sublinear relative to the width-linear asymptote; the qualitative divergence, not the exact exponent, is what Theorems 13 and 28 require, and we report the quantitative deficit as an open discrepancy.

Headline figure. Figure ?? (paired view: λ_ϕ and λ_θ on the same axes across the D sweep) shows the two eigenvalues diverging as D grows. The constancy of λ_θ is itself a strong qualitative finding: the spectral block carries no growing curvature with D, isolating the asymmetry to the spatial side.

7.3 Experiment 1.2: Two-timescale training dynamics

We train the compositional model under joint and frozen-spectral conditions across multiple capacities and architectures, logging the EGR every step. The predictions to be tested are Proposition 30 (the spectral gradient norm decays as $C/(1+\mu t)$ while the loss is fitted) and the consequence that the EGR collapse depth should scale monotonically with capacity.

Findings — CNN. On SpatialCNN at widths $D \in \{16, 64, 256, 1024\}$ with 3 seeds each:

- EGR collapse depth is monotonic in D:

$$\mathrm{EGR}^{\text{final}}_{D=16} \approx 0.40, \quad \mathrm{EGR}^{\text{final}}_{D=64} \approx 0.25, \quad \mathrm{EGR}^{\text{final}}_{D=256} \approx 0.10, \quad \mathrm{EGR}^{\text{final}}_{D=1024} \approx 0.08.$$

- Joint training final test loss diverges at high capacity. At $D = 1024$, joint test loss $= 2.27$ while frozen test loss $= 1.15$.
- Joint test accuracy peaks early (≈ 0.70) then decays as the model overfits to shortcuts. Frozen test accuracy is more stable.

This is the cleanest empirical validation of Theorem 28: on the canonical CNN architecture, the EGR exhibits exactly the capacity-monotonic collapse anticipated by Proposition 30 and the starvation regime of Theorem 28.

Findings – ViT. On SpatialViT, training enters a full memorization regime: train loss reaches 0.01–0.03 at all widths, while test loss is catastrophic (2.5–5.0). In this regime the EGR observable loses interpretive value: both gradient norms approach the noise floor and their ratio is dominated by stochasticity. The underlying prescription, however, still holds: frozen variants beat joint variants at every width tested.

Conclusion. Theorem 28's qualitative prediction (joint training fails by undertraining the spectral module) is universal: it manifests on both CNN and ViT spatial models. The specific EGR observable cleanly tracks the predicted decay in non-memorization regimes (CNN at our capacities) and becomes uninformative in the memorization regime (ViT). We discuss this limitation in Section 9.

7.4 Experiment 1.3: The equal-information killer test

The previous experiments validate the geometric (Theorem 13) and dynamical (Theorem 28) predictions on capacity-asymmetric compositional models. Experiment 1.3 tests a stronger qualitative claim suggested by the theory: *when spectral and spatial signals carry equal information about the label, end-to-end joint training will underperform freezing because of the shortcut dynamics.*

Calibration. We require an operational definition of "equal information" between the spectral pathway (per-pixel content of X along a direction $u \in \mathbb{R}^S$) and the spatial pathway (per-pixel content of X along an orthogonal direction $v \perp u$, with strength modulated by a smooth per-position function of the class). We compare three modes:

- *bayes*: scalar strengths α, β chosen such that a single-modality logistic regression on α-only data and a position-majority classifier on β-only data both achieve a target accuracy (0.75 in our experiments);

- *ntk*: α, β chosen such that the leading Gram-matrix eigenvalues of the per-pixel features from the two modalities are equal;

- *margin*: α, β chosen such that the L2 margins of single-modality logistic regressions are equal.

Setup. For each calibration mode we train the canonical CNN composition at width $D = 256$ for 150 epochs on $N_{\text{train}} = 512$ samples, repeated over 3 seeds. The four training conditions per mode are:

- *joint*: end-to-end training with both modules trainable;

- *frozen*: spectral module frozen at random initialization;

- *spectral-only*: $\beta = 0$, only spectral signal in X;

- *spatial-only*: $\alpha = 0$, only spatial signal in X.

Results. Under the `bayes` calibration we find:

- The single-modality baselines reach equal accuracy by construction: $\alpha = 0.82$, $\beta = 18.75$ produces spectral-only $= 0.746$, spatial-only $= 0.754$ (logistic regression and position-majority bounds respectively).

- Empirically on the full CNN model: spectral-only training (data with $\beta = 0$) achieves test accuracy 0.53, spatial-only training (data with $\alpha = 0$) achieves test accuracy 0.53. The CNN-achievable accuracies are equal, confirming the calibration is meaningful in architecture.

- Frozen-spectral training on the full data achieves 0.61, exploiting both modalities.

- **Joint training drops to** 0.48 – below both single-modality baselines (0.53) and 0.13 points below the frozen variant.

Two honest caveats bound this result. First, the effect size is modest in absolute terms: with 3 seeds and 128 test samples the per-condition standard error is roughly 0.02, so the 0.05 gap to the single-modality baselines is resolvable but not dramatic, and 0.48 is statistically indistinguishable from the 0.50 chance floor. Second, the CNN reaches only 0.53 on single-modality data against a 0.75 calibration target, so the network operates far from the Bayes bound and part of the joint-training deficit could in principle reflect generic overfitting with more trainable parameters. What the direction of the result nevertheless establishes is that joint training does not merely fail to combine the two modalities: it undershoots baselines that demonstrably use only one, which is consistent with active corruption of the spectral pathway rather than mere under-use, and is inconsistent with the "spectral features are simply irrelevant" reading (Remark 22). The frozen-vs-joint ordering, which is the theory's core prescription, holds in all three calibration modes (Table 1: +0.13, +0.04, +0.09); the undershoot below both single-modality baselines appears specifically in the `bayes` mode, where the modalities are matched most cleanly.

Comparison of calibration modes. Table 1 compares the three calibration modes on two criteria: (i) calibration quality, defined as $1 - |a_{\text{spec}} - a_{\text{spat}}|$ where $a_{\text{spec}}, a_{\text{spat}}$ are the achieved single-modality accuracies; and (ii) shortcut strength, defined as $a_{\text{frozen}} - a_{\text{joint}}$.

mode	α, β	spec / spat	joint	frozen	shortcut
bayes	0.82, 18.75	0.75/0.75	0.48	0.61	$+$**0.13**
ntk	0.99, 4.91	0.78/0.58	0.53	0.57	$+0.04$
margin	5.00, 5.00	0.95/0.59	0.60	0.69	$+0.09$

Table 1: Calibration mode comparison for the equal-information killer test (CNN $D = 256$, 3 seeds). *spec / spat*: achievable single-modality accuracies. *shortcut* $:= a_{\text{frozen}} - a_{\text{joint}}$. The `bayes` mode produces both the cleanest modality equalization and the largest shortcut effect.

Conclusion. The `bayes` calibration delivers the qualitative claim: when spectral and spatial signals carry equal information about the label (measured by what each modality alone can provide), end-to-end joint training underperforms both single-modality baselines, while freezing the spectral module recovers information that joint training squanders. Subject to the effect-size caveats above, this is empirical support consistent with the spectral-shortcut mechanism suggested by Theorem 28.

7.5 Robustness

In supplementary experiments we vary the optimizer (Adam vs. SGD with momentum), the noise level (0.05 vs. 0.10), and the spectral input dimension S. The capacity-driven joint-vs-frozen gap and the EGR collapse pattern hold across all variations, with two notable modulations: SGD makes the EGR signal cleaner (Adam normalizes per-parameter and partially masks the natural gradient ratio), and lower noise allows cross-entropy to fully saturate, accelerating the emergence of the shortcut.

[TODO: Phase 5, W15: write the robustness analyses into the supplement once Exp 1.3 is finalized.]

8 Real-Data Validation

The synthetic experiments in Section 7 verify the theorems and the EGR diagnostic in a controlled setting. We here briefly confirm that the predicted phenomenology manifests in a production-scale spectral-spatial classification task: per-pixel 4-class tissue classification on FTIR/QCL infrared hyperspectral images of breast cancer tissue microarrays. A companion empirical paper Dziuba et al. [2026] provides comprehensive ablations on this dataset and on a parallel prostate cancer dataset.

8.1 Setup

The data are 336×336 pixel cores collected by QCL spectroscopy over the 1000–1800 cm^{-1} fingerprint region. Each pixel carries an $S = 942$ dimensional spectrum (after stacking raw absorbance and its first two derivatives). The classification target is per-pixel tissue type. The compositional model is the Block Vision Transformer of O'Leary et al. O'Leary et al. [2026]: a linear per-pixel $f_{\boldsymbol{\theta}} : \mathbb{R}^{942} \to \mathbb{R}^{K}$, then 16×16 patch tokenization, a 12-layer / 12-head Vision Transformer of embedding width $M = 192$, a transposed-convolution decoder, and a convolutional segmentation head. Parameter counts obtained directly from the instantiated model are given in Table 2.

Channel convention. O'Leary et al.'s original BlockViT operates on the second-derivative spectrum alone (a single channel, $S = 965$ for their prostate FTIR data)—the field-standard preprocessing in infrared histopathology. The companion benchmark, and hence the checkpoints analyzed here, instead stacks raw absorbance with its first two derivatives (three channels,

Table 2: Module capacities of the BlockViT-v2 pipeline, counted from the instantiated model. K is the spectral bottleneck width; the two rows are the two configurations used in Section 8.

Configuration	C_f	C_g	C_g/C_f
$K = 64$ (end-to-end baseline)	61,108	18,271,540	299
$K = 128$ (matched to frozen variants)	121,588	21,472,564	177

$S = 942$). The choice affects only C_f: instantiating the same model on the single-channel field-standard input gives $C_f = 20{,}916$ and $C_g/C_f \approx 874$. The three-channel convention is therefore the *conservative* choice for this paper's purposes—under the field standard, the capacity asymmetry we analyze is roughly three times larger.

The capacity ratio is thus roughly two orders of magnitude (C_g/C_f between 177 and 299 depending on bottleneck width), and the total parameter count of the $K = 64$ configuration, 18.3×10^6, matches the figure reported for this architecture in the companion work. The spatial module's dominant blocks—the transformer (5.34×10^6) and the transposed convolution (9.44×10^6)—are both $\Theta(M^2)$ in the embedding width, so this architecture falls in the class for which Theorem 13 takes its parameter-count form: it predicts a module curvature disparity $\mathcal{D}_{\text{curv}} \gtrsim c_1' \sqrt{C_g/C_f}$, i.e. of order 13–17 times the constant c_1' of Lemmas 15–19; equivalently, in the theorem's native width variable, $\mathcal{D}_{\text{curv}} \gtrsim c_1 M$ with $M = 192$. Experiment 1.8 (Section 7) measures $\mathcal{D}_{\text{curv}}$ on this pipeline directly.

We evaluate five model variants under identical 5-fold cross-validation splits: end-to-end learned linear (joint), frozen random spectral, frozen PCA-128 spectral, frozen pretrained spectral encoder (pretrained on the pixel-level classification task before downstream training), and fine-tuned pretrained spectral (initialize from the pretrained encoder, then continue with end-to-end training).

8.2 Results

The mean test F1 scores across the 5 folds are summarized in Table 3.

Variant	Test F1
End-to-end learned linear (joint)	0.675
Frozen random spectral	0.70
Frozen PCA-128 spectral	0.78
Frozen pretrained spectral (Peak Windows)	0.84
Frozen pretrained spectral (MLP)	0.90
Frozen pretrained spectral (Sliding Win)	0.95
Fine-tuned pretrained spectral	0.79

Table 3: Test F1 on held-out FTIR/QCL breast cancer tissue cores (5-fold mean). End-to-end learned linear collapses to 0.675, while frozen random features outperform it ($0.70 > 0.675$). Frozen pretrained variants achieve 0.84–0.95. Fine-tuning the pretrained encoder degrades performance from 0.90 to 0.79.

Three observations align directly with the theory:

1. **Joint training loses to frozen-random.** End-to-end learned linear achieves 0.675, while frozen random spectral features achieve 0.70. This is the strongest single datapoint for Theorem 28: even uninformative features, when frozen, outperform supervised learning of the same spectral module.

2. **Frozen PCA improves over frozen random.** Adding informativeness (PCA) on top of stability (frozen) improves test F1 by 0.08. The benefit is monotone in feature informativeness, and the gap is preserved because the spatial model receives a non-moving target.

3. **Fine-tuning the pretrained encoder degrades it.** Starting from a strong frozen pretrained (0.90) and allowing the gradient to flow through it for fine-tuning drops performance to 0.79. This is the predicted dynamical consequence of Theorem 28: the same shortcut dynamics that prevent end-to-end training from succeeding also actively degrade pretrained features when they are allowed to update.

The ordering of variants is, top to bottom in Table 3, exactly what the theory predicts: any frozen variant beats any joint or fine-tuned variant; informativeness of the frozen features sets the final achievable level; joint and fine-tuned training is uniformly worse.

8.3 Diagnostic on real data

[TODO: Phase 6, W16-17. Compute EGR trajectories for each of the 5 variants during the original 5-fold training runs. Predict: learned linear shows early EGR collapse, frozen variants have undefined EGR (theta has no gradient), fine-tuned shows mid-training EGR collapse as the pretrained features are dragged off the manifold. Plot in a single figure with all variants overlaid.]

The companion empirical paper Dziuba et al. [2026] provides additional ablations: capacity ratio sweeps, prostate cancer cohort replication, train/val/test split sensitivity analysis. The numerical results across these ablations all conform to the predicted theory.

9 Discussion

9.1 Implications for practice

The theorems and experiments establish a concrete prescription. In any compositional spectral-spatial pipeline with substantial capacity asymmetry $C_g \gg C_f$, end-to-end joint training is expected to fail in a specific and predictable way: the spectral module stays near initialization while the spatial module fits the data using shortcuts in the unstructured feature map. Three operational responses follow.

Freeze the spectral module. The most direct response to Theorem 28 is to remove $\boldsymbol{\theta}$ from the optimization. Without a spectral update, the timescale separation is eliminated by construction, and $g_{\boldsymbol{\phi}}$ optimizes a standard supervised learning problem on fixed input features. If a pretrained spectral encoder is available (from unsupervised pretraining, self-supervised pretraining, or pre-task pretraining on the same data), it should be used frozen.

Use a fixed baseline. When pretraining is unavailable, a fixed but informative baseline $h(X)$ (e.g. PCA, fixed wavelet basis, random projection) can be injected as an additive component $Z = f_{\boldsymbol{\theta}}(X) + h(X)$. The fixed component ensures the spatial model receives a stable input even if $f_{\boldsymbol{\theta}}$ drifts; the additive structure allows fine-grained refinement of $f_{\boldsymbol{\theta}}$ without exposing the spatial model to the moving-target dynamics that produce the shortcut. We have not empirically tested this prescription in this paper; it follows directly from the theory and is a natural focus for follow-up work.

Track EGR. The EGR diagnostic detects the onset of starvation in real time. Practitioners should log it alongside training and validation loss. When $\mathrm{EGR}(t)$ drops below a chosen threshold (we recommend $0.1 \cdot \mathrm{EGR}(0)$), the theoretical analysis predicts that subsequent $\boldsymbol{\theta}$ updates are essentially noise: the spectral module has been absorbed into a spurious stationary point. At this point, freezing $\boldsymbol{\theta}$ and continuing to train $\boldsymbol{\phi}$ retains the run's investment while avoiding the negative consequences of further joint updates.

9.2 Limitations

We make four limitations explicit.

Linear spectral reduction. Theorems 13 and 28 are proven under the assumption that $f_{\boldsymbol{\theta}}$ is linear in $\boldsymbol{\theta}$. In practice some pipelines use shallow nonlinear spectral modules (1–2 layer MLPs). The intuition behind both theorems extends: nonlinear shallow $f_{\boldsymbol{\theta}}$ retains the property that the spectral block carries substantially less capacity than the spatial block, so the qualitative timescale separation persists. We verify empirically in Section 7 that the joint-vs-frozen gap remains under nonlinear $f_{\boldsymbol{\theta}}$. A formal proof for nonlinear $f_{\boldsymbol{\theta}}$ is left to future work.

Finite-width scaling exponent. The empirical scaling exponent of λ_ϕ vs. C_g is 0.70 ± 0.05 at the widths we tested, while the width-linear prediction of Lemma 15 for the tested architecture family is 1.0 (the SpatialMLP has $C_g = \Theta(M)$, so slope 1.0 applies against C_g as well as against M). The measurement is sublinear—it undershoots the asymptote. Whether the asymptote is approached at larger widths, or the deficit reflects the measurement and idealization gaps discussed in Section 7, is left open. Two consequences for the paper: the Theorem 13 bound should be read as a lower bound that may be loose at finite scale, and the empirical predictions should be interpreted as qualitative (monotonic growth) rather than tight (constant proportionality).

Memorization regime. For sufficiently overparametrized spatial models (ViT in our experiments), joint training enters a memorization regime in which train loss falls to near zero and both gradient norms approach the noise floor. In this regime the EGR observable becomes uninformative, even though the underlying prescription (freeze) continues to apply. Practitioners running highly overparametrized models should observe the symptoms (test loss divergence, train loss saturation) rather than relying on EGR alone.

Single domain validation. Our real-data validation is restricted to FTIR/QCL tissue classification in breast cancer. We expect the same mechanism to operate in hyperspectral remote sensing, Raman tissue classification, and mass spectrometry imaging, but those validations are future work.

9.3 Defense against the "linear triviality" critique

A natural reviewer objection is that, because $f_{\boldsymbol{\theta}}$ is linear, the analysis must reduce to principal component analysis. We address this in Remark 23 and emphasize again: PCA finds directions of maximum input variance independently of labels. Theorem 13 concerns curvature of the supervised cross-entropy loss, which depends on the current state of $g_{\boldsymbol{\phi}}$ through the Jacobian $\partial\hat{y}/\partial Z$. The pathology we identify is that this supervised signal fails to reach $\boldsymbol{\theta}$. A PCA-decomposed $f_{\boldsymbol{\theta}}$ would not exhibit the joint-vs-frozen gap we observe: indeed, our experiments include a frozen-PCA baseline that performs well, exactly because PCA decomposition does not require the supervised signal to flow through it.

9.4 The Spatial Dominance Conjecture: an open problem

A stronger statement than Theorem 28 is suggested by our experiments but not proven: *when spectral and spatial features carry equal information about the label, joint training exhibits an implicit bias toward the spatial pathway.* Pezeshki et al. Pezeshki et al. [2021] prove a related but distinct result in the NTK regime; Huang et al. Huang et al. [2022] prove a related result for explicit dual-modality networks but not for compositional single-modality pipelines. A proof of the spatial-dominance conjecture in our setting would require formalizing "equal information" in an operationally meaningful way (Shannon mutual information is bijection- invariant and therefore not the right primitive for gradient dynamics) and combining the NTK eigenvalue gap with our compositional Hessian analysis. We leave this as an open problem.

9.5 Beyond spectroscopy: where else this matters

Although our motivating application is infrared tissue classification, the mechanism we identify applies to any compositional architecture with capacity asymmetry between sequential modules. We highlight three areas where the same pathology likely operates.

Hyperspectral remote sensing. The dominant paradigm for hyperspectral image classification is a shallow spectral reduction (PCA, autoencoder, or 1D conv) feeding into a deep spatial classifier (3D CNN, ViT). The capacity asymmetry is typically large. Joint training is the de facto standard. Our analysis predicts the same shortcut emergence; whether the field has noticed it under the guise of "classification works without much spectral processing" is a question worth examining.

Multimodal foundation models. Vision-language and audio-language models compose a sequence of modality-specific encoders into a shared representation space. The modality with the larger encoder typically dominates the shared representation. While this has been documented empirically (Wang et al. Wang et al. [2020], Peng et al. Peng et al. [2022]), the mechanism is the same as ours: capacity-induced timescale separation combined with cross-entropy saturation. Our framework extends their empirical observations into the single-input compositional setting.

Video models with frozen image encoders. A standard practice in video understanding is to freeze a pretrained image encoder and train a temporal model on top. This is widely treated as "transfer learning," but the same mechanism we describe predicts that joint training would fail in a specific way – the image encoder would drift, the temporal model would absorb the shortcut, generalization would suffer. The empirical literature strongly supports the freezing practice; our analysis provides a mechanistic reason.

9.6 Concluding remarks

We have provided the first rigorous explanation for the failure of end-to-end joint training in compositional spectral-spatial deep learning. The phenomenon is geometric (Theorem 13) and dynamical (Theorem 28), is detectable in real time via the EGR diagnostic, and has a direct prescriptive remedy (freezing or fixed-baseline injection). We expect the framework to extend beyond spectroscopy to any compositional architecture with capacity asymmetry, and to provide a principled foundation for the empirical practice of frozen feature extractors that the field has converged on by necessity rather than design.

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A Supplementary Material

This supplement provides the full proofs of Theorems 13 and 28, additional experiments referenced in the main text, and robustness checks.

A.1 Full proof of Theorem 13

We restate Theorem 13 for reference:

> *Under Assumptions 1, 4, and 6, together with the head hypothesis of Lemma 15, the module curvature disparity satisfies, in expectation over random Kaiming (variance-scaled Gaussian) initialization, $\mathcal{D}_{\text{curv}}^{(M)} \geq c_1 \cdot M - c_2$, and for architecture families with $C_g = \Theta(M^2)$, $\mathcal{D}_{\text{curv}} \geq c_1' \sqrt{C_g/C_f} - c_2$, for constants depending on the data distribution, the initialization scheme, the bottleneck dimension K, the input spectral dimension S, and the architecture class of g_ϕ—but not on M or C_g.*

The proof rests on two lemmas: (i) the spatial block's top eigenvalue scales linearly with width M (Lemma 15, proved by an explicit witness direction in the head weights, with Karakida et al.'s mean-field law as the sharp asymptotic benchmark); (ii) the spectral block's top eigenvalue is capped by a C_g-independent operator-norm bound (Lemma 19). Cauchy interlacing (Lemma 17) is retained as an auxiliary fact but is not load-bearing for the disparity bound. We prove each lemma in turn, then combine.

A.1.1 Proof of Lemma 15

We show $\lambda_{\max}(J_\phi^\top J_\phi) = \Omega(M)$ in expectation over random Kaiming (variance-scaled Gaussian) initialization, under the head hypothesis of Lemma 15: the classifier head of g_ϕ is a dense layer over penultimate features $h_n \in \mathbb{R}^{M_h}$ with $M_h = \Theta(M)$ whose softmax-weighted mean Gram has top eigenvalue $\Omega(M_h)$ at initialization. The proof has three steps: (1) context—Karakida et al.'s exact width-linear law for the top Fisher eigenvalue in their model class, which we use as the asymptotic benchmark but do *not* invoke directly; (2) an exact spectral lower bound on the softmax inner matrix at initialization; (3) an elementary explicit witness direction in the head weights that yields $\Omega(M)$ self-containedly. Throughout, Grams are normalized as empirical means over the $N := N_{\text{data}}HW$ pixel positions; the unnormalized (summed) Grams differ by the fixed factor N, which does not affect any statement in M.

Step 1: The width-linear law of Karakida et al. (context). Karakida, Akaho, and Amari Karakida et al. [2019] analyze fully connected networks of depth $L \geq 2$ with layer widths $M_\ell = \alpha_\ell M$, linear outputs, and variance-scaled Gaussian (Kaiming-type) initialization $W_{ij}^\ell \sim \mathcal{N}(0, \sigma_w^2/M_{\ell-1})$, $b_i^\ell \sim \mathcal{N}(0, \sigma_b^2)$. For the empirical Fisher information of the squared-loss model, $F = \frac{1}{T}\sum_{t=1}^T \sum_k \nabla f_k(x_t) \nabla f_k(x_t)^\top$ (the empirical-mean Gram of the output Jacobian), their Theorem 4 gives, in the limit $M \gg 1$,

$$\lambda_{\max}(F) \;=\; \alpha\left(\tfrac{T-1}{T}\,\kappa_2 + \tfrac{1}{T}\,\kappa_1\right) M \;=\; \Theta(M),$$

where $\kappa_1, \kappa_2 = \Theta(1)$ are macroscopic constants computed from the layer-wise signal-propagation recursions and $\alpha = \sum_\ell \alpha_\ell \alpha_{\ell-1}$. Their Theorem 1 simultaneously shows the *mean* eigenvalue is $m_\lambda = C\kappa_1/M = \mathcal{O}(1/M)$: as the network widens, the FIM spectrum becomes extremely skewed—almost all directions flatten while the top eigenvalue grows linearly in width. We record two cautions. First, Theorem 4 assumes *all* layer widths scale proportionally with M; it does not cover networks with width-independent input or output dimensions (such as a classifier reading a fixed bottleneck K, where its coefficient α degenerates), so it cannot be applied verbatim to our spatial modules. Second, both statements are asymptotic mean-field equalities with no finite-M rate. Steps 2–3 therefore prove the $\Omega(M)$ bound self-containedly; Karakida's law serves as the sharp benchmark within its model class, validated numerically in their Figure 1.

Step 2: The softmax inner matrix at initialization. Our loss is softmax cross-entropy, not squared loss, so the spatial Gauss–Newton block carries the softmax inner matrix

$$G_{\phi\phi} \;=\; \frac{1}{N}\sum_{n=1}^{N} J^{(n)\top} H_\tau^{(n)} J^{(n)}, \qquad H_\tau \;:=\; \operatorname{diag}(p) - pp^\top,$$

where $J^{(n)} := \partial\hat{y}^{(n)}/\partial\phi$ is the per-pixel logit Jacobian and $p = \operatorname{softmax}(\hat{y})$. For any $v \in \mathbb{R}^{N_{\text{cls}}}$, the quadratic form is a variance: $v^\top H_\tau v = \sum_i p_i v_i^2 - (\sum_i p_i v_i)^2 = \operatorname{Var}_{i\sim p}(v_i) \ge p_{\min}\sum_i (v_i - \bar{v}_p)^2 \ge p_{\min}\|P_\perp v\|^2$, where $p_{\min} := \min_i p_i$, $\bar{v}_p := \sum_i p_i v_i$, the last inequality uses that the unweighted mean minimizes the sum of squared deviations, and $P_\perp := I - \frac{1}{N_{\text{cls}}}\mathbf{1}\mathbf{1}^\top$ projects onto the mean-zero subspace $\mathbf{1}^\perp$. Hence

$$H_\tau \;\succeq\; p_{\min}\, P_\perp.$$

The bound is used per pixel in Step 3; we do not take a minimum over pixels, so no uniform-over-pixels lower bound on $p_{\min}$ is needed anywhere in the proof. (For orientation: at Kaiming initialization the logits are $\mathcal{O}(1)$ uniformly in M, so $p_{\min}^{(n)}$ is of order $1/N_{\text{cls}}$ for typical n.)

Step 3: An explicit witness direction. Write the classifier head of g_ϕ as $\hat{y}_n = Wh_n + b$ with penultimate features $h_n = h(x_n) \in \mathbb{R}^{M_h}$, $M_h = \Theta(M)$. A perturbation of the head weights is a matrix $V \in \mathbb{R}^{N_{\text{cls}}\times M_h}$; it is a unit vector in parameter space precisely when $\|V\|_F = 1$, under any vectorization convention, and the logit perturbation it induces at pixel n is Vh_n. Define the *softmax-weighted feature Gram*

$$S_h^p \;:=\; \frac{1}{N}\sum_{n=1}^{N} p_{\min}^{(n)}\, h_n h_n^\top, \qquad p_{\min}^{(n)} := \min_i p_i^{(n)}, \tag{11}$$

and let u be a top unit eigenvector of S_h^p. Fix a unit contrast $c \in \mathbb{R}^{N_{\text{cls}}}$ with $c \perp \mathbf{1}$ (possible since $N_{\text{cls}} \ge 2$); by Step 2, at every pixel

$$c^\top H_\tau^{(n)} c \;=\; \operatorname{Var}_{i\sim p^{(n)}}(c_i) \;\ge\; p_{\min}^{(n)}\,\|P_\perp c\|^2 \;=\; p_{\min}^{(n)}.$$

Take the rank-one witness $V := c\,u^\top$, for which $\|V\|_F = \|c\|\,\|u\| = 1$ and $Vh_n = (h_n^\top u)\,c$. Then

$$\lambda_{\max}(G_{\phi\phi}) \;\ge\; \frac{1}{N}\sum_{n=1}^{N}(Vh_n)^\top H_\tau^{(n)}(Vh_n) \;=\; \frac{1}{N}\sum_{n=1}^{N}(h_n^\top u)^2\, c^\top H_\tau^{(n)} c \;\ge\; u^\top S_h^p u \;=\; \lambda_{\max}(S_h^p). \tag{12}$$

Note what (12) is: a deterministic, hypothesis-free inequality—one identity and one application of Step 2—valid at every parameter value. All N pixels contribute, and no minimum over pixels is taken, so no union bound and no factor of N enters.

The head hypothesis of Lemma 15 is exactly $\lambda_{\max}(S_h^p) \ge q_h M_h$ with probability at least $1-\delta_h$ over initialization, for $q_h > 0$ and $\delta_h < 1$ independent of M. On that event $\lambda_{\max}(G_{\phi\phi}) \ge q_h M_h = \Omega(M)$; since $\lambda_{\max} \ge 0$ always,

$$\mathbb{E}[\lambda_{\max}(G_{\phi\phi})] \;\ge\; (1-\delta_h)\, q_h M_h \;=\; \Omega(M),$$

which is the claim.

On the head hypothesis, and its counterpart in Karakida et al. Write $S_h := N^{-1}\sum_n h_n h_n^\top$ for the unweighted mean feature Gram. The hypothesis $\lambda_{\max}(S_h^p) = \Omega(M_h)$ asks two things at once: that the penultimate features carry a common direction of $\Theta(M_h)$ energy across inputs

$(\lambda_{\max}(S_h) = \Omega(M_h))$, and that the initial predictions are non-degenerate on the pixels supplying that energy. The second is mild at initialization—the logits are $\mathcal{O}(1)$ uniformly in M, since the head weights have variance $\Theta(1/M_h)$ against $\|h_n\|^2 = \mathcal{O}(M_h)$, so $p_{\min}^{(n)}$ is bounded below by a constant of order $1/N_{\text{cls}}$ on all but a small fraction of pixels, and S_h^p dominates that constant times the unweighted Gram restricted to those pixels. (Proposition 35 below turns this sketch into a proof for a concrete head.) The first is the substantive condition. It is implied by $\|\bar{h}\|^2 \geq qM_h$ on the *mean* feature vector $\bar{h} := N^{-1}\sum_n h_n$: for $\bar{u} = \bar{h}/\|\bar{h}\|$,

$$\lambda_{\max}(S_h) \;\geq\; \bar{u}^\top S_h \bar{u} \;=\; \frac{1}{\|\bar{h}\|^2}\cdot\frac{1}{N}\sum_n (h_n^\top \bar{h})^2 \;\geq\; \frac{1}{\|\bar{h}\|^2}\left(\frac{1}{N}\sum_n h_n^\top \bar{h}\right)^2 \;=\; \frac{\|\bar{h}\|^4}{\|\bar{h}\|^2} \;=\; \|\bar{h}\|^2,$$

the middle step being Jensen's inequality (the mean of squares dominates the square of the mean). At variance-scaled initialization each entry of h_n has $\Theta(1)$ second moment (the signal-propagation recursions of Karakida et al. Karakida et al. [2019], Eq. (9)), so the condition holds whenever the activations are correlated across inputs—in particular for any nonnegative or nonzero-mean activation (ReLU and its variants), and for odd activations whenever the bias variance $\sigma_b^2 > 0$ or the inputs are themselves correlated.

Note that per-entry second moments alone are *not* enough: $\|h_n\|^2 = \Theta(M_h)$ for every n gives only $\operatorname{tr}(S_h) = \Theta(M_h)$, hence $\lambda_{\max}(S_h) \geq \Theta(M_h)/\min(N, M_h)$, which is weaker by a factor of N. Cross-input correlation, not per-vector norm, is what makes the top eigenvalue grow.

The same quantity carries the width-linear term in Karakida et al.'s Theorem 4, $\lambda_{\max} = \alpha(\frac{T-1}{T}\kappa_2 + \frac{1}{T}\kappa_1)M$: their κ_2 is built from the cross-input activation correlations $\hat{q}_{st}$, and their footnote notes that odd activations with $\sigma_b = 0$ give $\hat{q}_{st} = 0$, $\kappa_2 = 0$, leaving only the κ_1/T term. The parallel is close but not an identity: on such a family our hypothesis does not fail outright either, but survives only with $q_h = \Theta(1/N)$—mirroring their surviving $\alpha\kappa_1 M/T$ term. What both lose in the degenerate case is the sample-size-independent leading constant, not the width scaling itself. For CNN and ViT heads we state the hypothesis as an explicit assumption rather than a theorem, and check it empirically in Section 7.

Proposition 35 (A fully proved instance). *Fix bottleneck features $z_1, \ldots, z_N \in \mathbb{R}^K$ with $\|z_n\| \leq R$, and let the spatial module's head be the two-layer ReLU network*

$$h_n = \operatorname{ReLU}(W_1 z_n + b_1), \qquad \hat{y}_n = W_2 h_n + b_2,$$

with Kaiming Gaussian initialization: entries of $W_1 \in \mathbb{R}^{M_h \times K}$ i.i.d. $\mathcal{N}(0, \sigma_w^2/K)$, entries of $W_2 \in \mathbb{R}^{N_{\text{cls}} \times M_h}$ i.i.d. $\mathcal{N}(0, \sigma_w^2/M_h)$, biases i.i.d. $\mathcal{N}(0, \sigma_b^2)$ with $\sigma_b > 0$, and $N_{\text{cls}} \geq 2$. Then there exist constants $q_h > 0$, $M_0 < \infty$, and $C_J < \infty$, depending only on $\sigma_w, \sigma_b, R, K, N_{\text{cls}}$—not on M_h or N—such that for all $M_h \geq M_0$:

- *(i) (head hypothesis) $\lambda_{\max}(S_h^p) \geq q_h M_h$ with probability at least $1/2$ over the initialization; consequently $\mathbb{E}[\lambda_{\max}(G_{\phi\phi})] \geq \frac{1}{2}q_h M_h$;*
- *(ii) (input-Jacobian cap at initialization) $\|\partial\hat{y}/\partial z\|_{\text{op}} \leq C_J$ with probability at least $3/4$, uniformly in M_h.*

Both hypotheses of Theorem 13 therefore hold, at initialization, for this architecture with no unproved assumptions.

Proof. Step A (mean feature vector). Write $g_{n,i} = w_i^\top z_n + b_{1,i}$ for the pre-activations, so $g_{n,i} \sim \mathcal{N}(0, s_n^2)$ with $s_n^2 = \sigma_w^2\|z_n\|^2/K + \sigma_b^2 \in [\sigma_b^2, s_{\max}^2]$, $s_{\max}^2 := \sigma_w^2 R^2/K + \sigma_b^2$. Since $\mathbb{E}\operatorname{ReLU}(\mathcal{N}(0, s^2)) = s/\sqrt{2\pi}$, each coordinate of the mean feature vector $\bar{h} = N^{-1}\sum_n h_n$ has expectation

$$m \;:=\; \mathbb{E}[\bar{h}_i] \;=\; \frac{1}{N}\sum_n \frac{s_n}{\sqrt{2\pi}} \;\geq\; \frac{\sigma_b}{\sqrt{2\pi}} \;>\; 0.$$

The rows $(w_i, b_{1,i})$ are i.i.d. across i, so the coordinates $\bar{h}_i$ are i.i.d. with $\mathbb{E}[\bar{h}_i^2] \le N^{-1}\sum_n \mathbb{E}[\mathrm{ReLU}(g_{n,i})^2] \le s_{\max}^2$ and (being averages of sub-Gaussian variables) finite fourth moment bounded by a constant $C_4(s_{\max})$. By Chebyshev's inequality applied to the i.i.d. sum $\|\bar{h}\|^2 = \sum_i \bar{h}_i^2$,

$$\Pr\Big[\|\bar{h}\|^2 < \tfrac{1}{2}m^2 M_h\Big] \;\le\; \frac{4\,C_4(s_{\max})}{m^4\,M_h} \;\xrightarrow[M_h\to\infty]{}\; 0.$$

On the complementary event, the Jensen step above gives $\lambda_{\max}(S_h) \ge \|\bar{h}\|^2 \ge \frac{1}{2}m^2 M_h$, and moreover the witness direction $\bar{u} = \bar{h}/\|\bar{h}\|$ certifies $A := N^{-1}\sum_n (h_n^\top \bar{u})^2 \ge \frac{1}{2}m^2 M_h$.

Step B (softmax weights). Condition on $\mathcal{F}_1 = \sigma(W_1, b_1)$, which fixes the features h_n, the direction $\bar{u}$, and the energies $a_n := (h_n^\top \bar{u})^2$; the head randomness (W_2, b_2) is independent of $\mathcal{F}_1$. Call pixel n *tame* if $\|h_n\|^2 \le 2s_{\max}^2 M_h$. For a tame pixel, the logits $\hat{y}_n \mid \mathcal{F}_1$ have i.i.d. Gaussian entries with variance $\sigma_w^2\|h_n\|^2/M_h + \sigma_b^2 \le 2\sigma_w^2 s_{\max}^2 + \sigma_b^2 =: s_*^2$, so by a Gaussian tail bound on the logit range, there are constants $c_p(s_*, N_{\mathrm{cls}}) > 0$ and $\delta_0 \le 1/16$ with $\Pr[p_{\min}^{(n)} < c_p \mid \mathcal{F}_1] \le \delta_0$ for every tame pixel. Untamed pixels are exponentially rare: $\|h_n\|^2/M_h$ is an average of M_h i.i.d. sub-exponential variables with mean $\le s_{\max}^2/2$ (Bernstein's inequality; Vershynin Vershynin [2018], Thm. 2.8.1), so $\mathbb{E}\big[N^{-1}\sum_n a_n \mathbf{1}[n \text{ untame}]\big] \le \mathbb{E}\Big[N^{-1}\sum_n \|h_n\|^2\,\mathbf{1}[\|h_n\|^2 > 2s_{\max}^2 M_h]\Big] \le \varepsilon(M_h)M_h$ with $\varepsilon(M_h) \to 0$ exponentially.

Step C (assembly). Let $B \subseteq \{1,\dots,N\}$ be the pixels with $p_{\min}^{(n)} < c_p$ or untame geometry. Combining Step B, $\mathbb{E}[\,N^{-1}\sum_{n\in B} a_n \mid \mathcal{F}_1] \le \delta_0 A + \varepsilon(M_h)M_h$, and for M_h large enough that $\varepsilon(M_h) \le m^2\delta_0$, this is at most $3\delta_0 A \le \frac{3}{16}A$. By Markov's inequality, $\Pr[\,N^{-1}\sum_{n\in B} a_n > \frac{1}{2}A \mid \mathcal{F}_1\,] \le 3/8$. On the complementary event,

$$\lambda_{\max}(S_h^p) \;\ge\; \bar{u}^\top S_h^p \bar{u} \;\ge\; c_p \cdot \frac{1}{N}\sum_{n\notin B} a_n \;\ge\; \frac{c_p}{2}\,A \;\ge\; \frac{c_p m^2}{4}\,M_h.$$

Adding the failure probabilities (Step A's $O(1/M_h)$ plus 3/8) keeps the total below 1/2 for $M_h \ge M_0$; take $q_h := c_p m^2/4$. The in-expectation clause follows from $\lambda_{\max} \ge 0$ and (12). This proves (i).

Step D (input-Jacobian cap). At any input z, the Jacobian is $\partial\hat{y}/\partial z = W_2 D_z W_1$ with $D_z = \mathrm{diag}(\mathbf{1}[W_1 z + b_1 > 0])$. Conditioning on (W_1, b_1) and using that the rows of W_2 are i.i.d. $\mathcal{N}(0, (\sigma_w^2/M_h)I)$,

$$\mathbb{E}\,\|W_2 D_z W_1\|_F^2 \;=\; N_{\mathrm{cls}} \cdot \frac{\sigma_w^2}{M_h} \cdot \mathbb{E}\,\|D_z W_1\|_F^2 \;\le\; N_{\mathrm{cls}} \cdot \frac{\sigma_w^2}{M_h} \cdot \mathbb{E}\,\|W_1\|_F^2 \;=\; N_{\mathrm{cls}}\,\sigma_w^4,$$

using $\mathbb{E}\,\|W_1\|_F^2 = M_h K \cdot \sigma_w^2/K = M_h \sigma_w^2$. Markov's inequality gives $\|\partial\hat{y}/\partial z\|_{\mathrm{op}} \le \|\cdot\|_F \le 2\sigma_w^2\sqrt{N_{\mathrm{cls}}} =: C_J$ with probability at least 3/4, uniformly in M_h. (Note the naive product bound $\|W_2\|_{\mathrm{op}}\,\|W_1\|_{\mathrm{op}}$ would grow like $\sqrt{M_h/K}$; the composition is $O(1)$ because the top directions of the two random factors are generically misaligned—which the Frobenius computation captures without any alignment analysis.) This proves (ii). $\square$

Remark 36. Proposition 35 is deliberately minimal: one hidden layer, ReLU, bias variance $\sigma_b^2 > 0$ (which guarantees activation correlation across inputs; cf. the κ_2 discussion above). Its role is to certify that Theorem 13's two structural hypotheses are *theorems*, not merely assumptions, for at least one concrete member of each side of the pipeline—so the disparity law has a fully verified instance. For deeper heads and attention-based modules the hypotheses revert to assumption status, argued by signal propagation and checked empirically in Section 7.

The other two conventions. The bound just proved is for the Gauss-Newton block $G_{\phi\phi} = J_\phi^\top H_\tau J_\phi$ of (6), which is the object appearing in Definition 11. Two neighbouring matrices inherit it. For the *logit* Gram, $H_\tau \preceq I$ gives $J_\phi^\top J_\phi \succeq G_{\phi\phi}$, so $\lambda_{\max}(J_\phi^\top J_\phi) \ge \lambda_{\max}(G_{\phi\phi}) = \Omega(M)$

immediately. For the *residual* Gram $J_{\boldsymbol{\phi}}^{\mathrm{res}\top} J_{\boldsymbol{\phi}}^{\mathrm{res}} = J_{\boldsymbol{\phi}}^{\top} H_\tau^2 J_{\boldsymbol{\phi}}$, note that H_τ is symmetric with $H_\tau \mathbf{1} = 0$, so $\mathbf{1}^\perp$ is invariant and, by Step 2, $H_\tau^{(n)} \succeq p_{\min}^{(n)} I$ there; squaring the eigenvalues on that subspace gives $(H_\tau^{(n)})^2 \succeq (p_{\min}^{(n)})^2 P_\perp$, so the identical witness argument runs with the weights $(p_{\min}^{(n)})^2$ in place of $p_{\min}^{(n)}$ and yields $\lambda_{\max}(J_{\boldsymbol{\phi}}^{\mathrm{res}\top} J_{\boldsymbol{\phi}}^{\mathrm{res}}) \geq \lambda_{\max}(S_h^{p^2}) = \Omega(M)$ under the corresponding head hypothesis. All three conventions therefore satisfy Lemma 15, differing only in the softmax weighting carried by the feature Gram. This completes the proof. $\square$

Remark 37 (Scope, and the role of the mean-field law). The witness argument uses only: (a) a dense classifier head over penultimate features of dimension $\Theta(M)$; (b) the head hypothesis $\lambda_{\max}(S_h^p) \geq q_h M_h$ at initialization—a *cross-input correlation* condition, which variance-scaled initialization does not by itself deliver (see the counterexample of per-vector norms above) but which holds for the standard activation choices discussed there; and (c) softmax cross-entropy with $N_{\mathrm{cls}} \geq 2$. Given (b), the inequality $\lambda_{\max}(G_{\boldsymbol{\phi}\boldsymbol{\phi}}) \geq \lambda_{\max}(S_h^p)$ is deterministic and architecture-agnostic, so the argument applies to per-pixel dense heads on MLP, CNN (channel count M), and ViT (embedding dimension M) spatial modules alike. For fully connected modules (b) follows from standard signal propagation at Kaiming initialization; for CNN and ViT heads we adopt it as an explicit assumption checked empirically. No i.i.d. Gaussian input assumption is required. Karakida et al.'s Theorem 4 plays the role of the sharp asymptotic benchmark: within its model class (fully connected, all widths proportional to M, squared loss), it upgrades the witness lower bound to an exact width-linear law with explicit constant, and their Theorem 1 supplies the complementary fact that the *mean* eigenvalue shrinks as $\mathcal{O}(1/M)$, so curvature concentrates in a vanishing fraction of directions as the module widens. The loss transfer from squared loss to cross-entropy is handled exactly at initialization by Step 2, at the cost of the constant $p_{\min}$.

Remark 38 (Empirical status). Karakida et al.'s own numerical validation (their Figure 1, right column) confirms $\lambda_{\max}$ growing linearly in M, matching the exact constant of their Theorem 4, for tanh, ReLU, and linear fully connected networks. Our Experiment 1.1 tests the law on the SpatialMLP family, whose parameter count is *linear* in width ($C_g = \Theta(M)$, since input and output dimensions are fixed), so the width-linear law predicts a log–log slope of 1.0 against C_g as well as against M. The measured slope is 0.70 ± 0.05: the qualitative claim of Lemma 15—λ_ϕ grows without bound while $\lambda_{\boldsymbol{\theta}}$ stays capped—is confirmed unambiguously, but the measured exponent is sublinear, short of the width-linear asymptote. Candidate causes (finite width, the loss-Hessian versus Gauss–Newton measurement gap, correlated non-Gaussian features) are discussed in Section 7. We report the discrepancy rather than explain it away.

A.1.2 Proof of Lemma 17

The claim that $\lambda_{\max}(M) \geq \max\{\lambda_{\max}(A), \lambda_{\max}(D)\}$ for a symmetric block matrix $M = \begin{pmatrix} A & B \\ B^\top & D \end{pmatrix}$ is a direct application of Cauchy's interlacing theorem to principal submatrices. See Bhatia Bhatia [1997], Corollary III.1.5.

For the eigenvector v achieving $\lambda_{\max}(A) = v^\top A v$, the extension $\tilde{v} = \begin{pmatrix} v \\ 0 \end{pmatrix}$ has $\tilde{v}^\top M \tilde{v} = v^\top A v$, so $\lambda_{\max}(M) \geq \tilde{v}^\top M \tilde{v} / \|\tilde{v}\|^2 = \lambda_{\max}(A)$. The same argument for D completes the proof. $\square$

A.1.3 Proof of Lemma 19

The proof factors the spectral Jacobian via the chain rule and bounds each factor's operator norm using Assumptions 1, 4, and 6. We bound the *logit* Jacobian $J_{\boldsymbol{\theta}} = \partial \hat{y} / \partial \boldsymbol{\theta}$ of (3); the Gauss-Newton block and the residual Jacobian then inherit the bound, since both insert factors of H_τ with $0 \preceq H_\tau \preceq I$ (Step 4).

Step 1: Chain-rule factorisation. With $Z = f_{\boldsymbol{\theta}}(X)$ and $\hat{y} = g_{\boldsymbol{\phi}}(Z)$, the chain rule gives, in the normalization of (3) (the $N^{-1/2}$ riding on the right-hand factor, as fixed in Section 3),

$$J_{\boldsymbol{\theta}} \;=\; J_Z \cdot \frac{\partial Z}{\partial \boldsymbol{\theta}}, \qquad J_Z := \partial g_{\boldsymbol{\phi}} / \partial Z,$$

where J_Z is unnormalized—the object Assumption 6 bounds. Since $g_{\boldsymbol{\phi}}$ acts on one image at a time, the dataset-stacked J_Z is block diagonal across images, so its operator norm is the maximum over per-image blocks and Assumption 6 applies to the stack with the same constant $\widetilde{L}$. (For the residual map $r(\boldsymbol{\theta}, \boldsymbol{\phi}) = \tau(g_{\boldsymbol{\phi}}(Z))$ with $\tau(\hat{y}) = \mathrm{softmax}(\hat{y}) - y_{\mathrm{onehot}}$, the same computation carries the extra left factor $J_\tau := \partial\tau/\partial\hat{y} = \mathrm{diag}(p) - pp^\top = H_\tau$, the softmax Jacobian, which satisfies $\|H_\tau\|_{\mathrm{op}} \le 1$; see Botev, Ritter, and Barber Botev et al. [2017].)

Step 2: Operator-norm inequality. Sub-multiplicativity of the operator norm gives

$$\|J_{\boldsymbol{\theta}}\|_{\mathrm{op}} \;\le\; \|J_Z\|_{\mathrm{op}} \cdot \left\|\frac{\partial Z}{\partial \boldsymbol{\theta}}\right\|_{\mathrm{op}} \;\le\; \widetilde{L} \cdot \left\|\frac{\partial Z}{\partial \boldsymbol{\theta}}\right\|_{\mathrm{op}},$$

using Assumption 6 on the first factor.

Step 3: The input-Jacobian factor, exactly. Under Assumption 1, $Z_n = W^\top x_n$ with $\boldsymbol{\theta} = \mathrm{vec}(W)$, $W \in \mathbb{R}^{S \times K}$. The per-pixel Jacobian is $\partial Z_n / \partial \mathrm{vec}(W) = I_K \otimes x_n^\top$ (Magnus and Neudecker Magnus and Neudecker [1988], Ch. 9). Stacking over the N pixel positions with the $N^{-1/2}$ normalization of (3) gives $\partial Z / \partial \boldsymbol{\theta} \in \mathbb{R}^{NK \times SK}$ with

$$\left(\frac{\partial Z}{\partial \boldsymbol{\theta}}\right)^\top \frac{\partial Z}{\partial \boldsymbol{\theta}} \;=\; \frac{1}{N} \sum_{n=1}^{N} (I_K \otimes x_n) \left(I_K \otimes x_n^\top\right) \;=\; I_K \otimes \Sigma_X,$$

hence, *exactly*,

$$\left\|\frac{\partial Z}{\partial \boldsymbol{\theta}}\right\|_{\mathrm{op}}^2 \;=\; \lambda_{\max}(I_K \otimes \Sigma_X) \;=\; \lambda_{\max}(\Sigma_X),$$

with Σ_X the mean per-pixel input Gram of (1); Assumption 4 guarantees $\lambda_{\max}(\Sigma_X) < \infty$ with high probability, with a limit independent of N. Note that no factor of K appears: the Kronecker structure replicates the same spectrum K times rather than accumulating it.

Step 4: Assembling. Combining Steps 2 and 3,

$$\|J_{\boldsymbol{\theta}}\|_{\mathrm{op}}^2 \;\le\; \widetilde{L}^2 \cdot \lambda_{\max}(\Sigma_X) \;=:\; C_{\boldsymbol{\theta}}.$$

For the Gauss-Newton block, $G_{\boldsymbol{\theta}\boldsymbol{\theta}} = J_{\boldsymbol{\theta}}^\top H_\tau J_{\boldsymbol{\theta}} \preceq J_{\boldsymbol{\theta}}^\top J_{\boldsymbol{\theta}}$ since $H_\tau \preceq I$, so $\lambda_{\max}(G_{\boldsymbol{\theta}\boldsymbol{\theta}}) \le C_{\boldsymbol{\theta}}$; for the residual Jacobian, $\|H_\tau J_{\boldsymbol{\theta}}\|_{\mathrm{op}} \le \|J_{\boldsymbol{\theta}}\|_{\mathrm{op}}$ gives the same cap. The constant $C_{\boldsymbol{\theta}}$ depends on the classifier's input Lipschitz constant $\widetilde{L}$ (which may grow with the depth of $g_{\boldsymbol{\phi}}$ but not with its width M; see the discussion of Assumption 6) and the top eigenvalue of the mean input Gram $\lambda_{\max}(\Sigma_X)$. It does not depend on M, on the spatial parameter count C_g, or on the dataset size N. This completes the proof. ☐

Remark 39 (Why an operator-norm cap suffices). An alternative route via the smallest positive eigenvalue $\lambda_{\min}^+(J_{\boldsymbol{\theta}}^\top J_{\boldsymbol{\theta}})$ requires either an exact Kronecker factorisation (valid only for per-pixel classifiers) or a trace-over-rank argument via the softmax residual dimension—both of which need additional structural assumptions on $g_{\boldsymbol{\phi}}$. The operator-norm bound above is significantly simpler, applies to general nonlinear spatial classifiers (CNN, ViT), and is exactly what Theorem 13 needs: a C_g-independent cap on the spectral block's *top* eigenvalue, so that $\mathcal{D}_{\mathrm{curv}} = \lambda_{\max}(G_{\boldsymbol{\phi}\boldsymbol{\phi}}) / \lambda_{\max}(G_{\boldsymbol{\theta}\boldsymbol{\theta}}) \ge \Omega(M)/C_{\boldsymbol{\theta}}$.

Remark 40 (Softmax Jacobian bound). The bound $\|\mathrm{diag}(p)-pp^\top\|_{\mathrm{op}} \le 1$ is standard: $\mathrm{diag}(p)-pp^\top$ is the covariance matrix of the categorical distribution with parameter p (equivalently, the Jacobian of the softmax map), hence positive semi-definite with $\lambda_{\max} \le \max_i p_i \le 1$. See Botev, Ritter, and Barber Botev et al. [2017] for its role in Gauss–Newton computations.

A.1.4 Combining the lemmas: full proof of Theorem 13

Assemble Lemmas 15 and 19 to complete the proof. By Lemma 15, in expectation over random Kaiming (variance-scaled Gaussian) initialization,

$$\lambda_{\max}(G^{(M)}_{\boldsymbol{\phi}\boldsymbol{\phi}}) \;\ge\; c_\phi \cdot M$$

for a constant $c_\phi > 0$ bounded below independently of M: the witness bound (12) gives $\mathbb{E}[\lambda_{\max}] \ge (1-\delta_h)\, q_h M_h \ge (1-\delta_h)\, q_h \beta M$, where $\beta > 0$ is the width-ratio floor $M_h \ge \beta M$ supplied by the head hypothesis, so one may take $c_\phi := (1-\delta_h) q_h \beta$. The witness route delivers this with no additive correction; within Karakida et al.'s Karakida et al. [2019] proportional-width model class, their Theorem 4 sharpens the width-linear rate to an exact asymptotic law. By Lemma 19,

$$\lambda_{\max}(G_{\boldsymbol{\theta}\boldsymbol{\theta}}) \;\le\; \|J_{\boldsymbol{\theta}}\|^2_{\mathrm{op}} \;\le\; C_{\boldsymbol{\theta}}$$

with $C_{\boldsymbol{\theta}}$ independent of M and C_g. Dividing,

$$\mathcal{D}^{(M)}_{\mathrm{curv}} \;=\; \frac{\lambda_{\max}(G^{(M)}_{\boldsymbol{\phi}\boldsymbol{\phi}})}{\lambda_{\max}(G_{\boldsymbol{\theta}\boldsymbol{\theta}})} \;\ge\; \frac{c_\phi \cdot M}{C_{\boldsymbol{\theta}}} \;=\; c_1 \cdot M, \tag{13}$$

where $c_1 := c_\phi / C_{\boldsymbol{\theta}}$ depends only on the initialization scheme, the architecture class of $g_{\boldsymbol{\phi}}$, the classifier's input Lipschitz constant $\widetilde{L}$, the bottleneck dimension K, the input spectral dimension S, and the data distribution—exactly the dependencies claimed in the theorem statement. The witness route needs no additive correction ($c_2 = 0$ suffices); the subtracted constant is retained in the statement so that sharper asymptotic constants, which hold only for M large, may also be substituted. Under $C_g = \Theta(M^2)$, $M = \Theta(\sqrt{C_g})$ and the ratio form $\mathcal{D}_{\mathrm{curv}} \ge c_1' \sqrt{C_g/C_f} - c_2$ follows on multiplying and dividing by $\sqrt{C_f}$. This completes the proof. $\square$

Remark 41 (What Theorem 13 does and does not claim). Theorem 13 is a scaling law for the curvature ratio between two Gauss-Newton blocks; it does not claim that the full joint Gauss-Newton matrix has a large condition number in the strict $\lambda_{\max}/\lambda_{\min}$ sense (which is trivially infinite under rank-deficiency), nor that joint gradient descent produces disparate effective learning rates (which is the separate content of Theorem 28, under its own hypotheses). The scaling $\mathcal{D}_{\mathrm{curv}} = \Omega(M)$ (equivalently $\Omega(\sqrt{C_g/C_f})$ for $\Theta(M^2)$-parameter families) is a purely geometric statement about how curvature concentrates in the downstream module as the spatial network widens.

Remark 42 (Width scaling and the empirical exponent). The $\sqrt{C_g}$ scaling reflects the fact that Karakida's underlying result Karakida et al. [2019] (Theorem 4) gives $\lambda_{\max}$ in terms of the network's characteristic *width* M, not its total parameter count. For a fixed-depth architecture with $C_g = \Theta(M^2)$, the equivalent scaling is $\lambda_{\max}(J_{\boldsymbol{\phi}}^\top J_{\boldsymbol{\phi}}) \sim \sqrt{C_g}$ (log–log slope 0.5 against C_g). Experiment 1.1's SpatialMLP is *not* in this class: its parameter count is linear in width ($C_g = \Theta(M)$), so the applicable prediction there is slope 1.0 against C_g, and the measured 0.70 ± 0.05 is sublinear (Remark 38). The two parameter-count forms must not be conflated; the theorem's native variable is M.

A.2 Supplementary material for Theorem 28

The main-text proof of Theorem 28 is complete for the gradient-flow model (7): the displacement bound follows by direct integration of the residual envelope, with the operator-norm cap supplied by Lemma 19. The supplementary work concerns the hypotheses and the discrete extension.

[TODO: Remaining supplement work for Theorem 28: (i) discrete-SGD extension via Robbins–Monro tracking (Borkar Borkar [2008], Ch. 2, Thm. 2), with gradient noise entering the constants; (ii) informal NTK derivation of the shortcut fitting rate $\mu(C_g)$ for the linearized fast subsystem, adapting Pezeshki et al. Pezeshki et al. [2021]; (iii) empirical verification that measured residual trajectories on the synthetic problems are enveloped by $C/(1+\mu t)$, with fitted μ increasing in D (data available from Experiment 1.2 logs).]

A.3 Empirical robustness

We here report robustness experiments referenced in the main text.

A.3.1 Optimizer ablation (SGD vs Adam)

[TODO: Phase 5 follow-up. Show EGR collapse pattern in both regimes; note Adam masks the natural gradient ratio.]

A.3.2 Loss-variant ablation

[TODO: Phase 5 follow-up. Cross-entropy, focal loss, label smoothing.]

A.3.3 Regularizer ablation

[TODO: Phase 5 follow-up. Weight decay, dropout, batch norm. Pezeshki et al. Pezeshki et al. [2021], App. B, already show gradient starvation is robust to these; we reproduce briefly in our setting.]

A.3.4 Smallest capacity ratio where the pathology disappears

[TODO: Phase 5 follow-up. Sweep C_g/C_f from 1 to 1000, identify the threshold below which joint training reaches frozen performance.]